

1 Main thread of the algorithm

The full cutting algorithm is quite complicated. However, it contains a main thread that is much simpler, and many of the complications actually come from the additional structures and conditions that have to be added to this main thread. In this section, we present the proof in a much simplified scenario, which basically isolates this main thread. The additions needed in full generality, and the reduction to this simplified case, will be discussed in Section 1.5.

It is useful to keep in mind the following facts and conventions about molecules:

- We often refer to $\mathbb{M}' \subseteq \mathbb{M}$ as a (sub-)molecule by regarding any bond between $\mathbb{M}'$ and $\mathbb{M} \setminus \mathbb{M}'$ as a *free end* in $\mathbb{M}'$. Also, once we cut out any elementary component S , we will ignore it for the rest of the cutting algorithm and consider the molecule $\mathbb{M} \setminus S$ only.
- An atom $\mathbf{m}$ is a *parent* (resp. *child*) of another atom $\mathbf{n}$ if $\mathbf{m}$ is connected to $\mathbf{n}$ by a bond with $\mathbf{m}$ being above (resp. below) $\mathbf{n}$. We say $\mathbf{m}$ is a *descendant* (resp. *ancestor*) of $\mathbf{n}$ if $\mathbf{m}$ is obtained by iteratively taking children (resp. parents) starting from $\mathbf{n}$. Note that this is a monotonic process, i.e. one cannot get back to the start by iteratively taking parents or iteratively taking children.
- If we cut any atom $\mathbf{n}$ as free, then this introduces a *bottom fixed end* at each *parent* of $\mathbf{n}$ (if exists), and a *top fixed end* at each *child* of $\mathbf{n}$ (if exists). Recall also that each atom has at most 2 parents and at most 2 children.

We start by defining the simplified scenario to be studied in this section.

Definition 1.1. We say $\mathbb{M}$ is a *simplified 2-layer model*, if

- (1) $\mathbb{M}$ has no O-atoms, and we ignore all the degeneracies;
- (2) $\mathbb{M}$ is a 2-layer molecule $\mathbb{M} = \mathbb{M}_U \cup \mathbb{M}_D$ with an “up layer” $\mathbb{M}_U$ and a “down layer” $\mathbb{M}_D$, and no atom in $\mathbb{M}_D$ is parent of an atom in $\mathbb{M}_U$;
- (3) Each of $\mathbb{M}_U$ and $\mathbb{M}_D$ is a tree, has no fixed end, and the number of bonds connecting an atom in $\mathbb{M}_U$ to an atom in $\mathbb{M}_D$, is at least ρ (where ρ is a fixed parameter; in the main nontrivial case we have $\rho \gg 1$).

The goal is to cut $\mathbb{M}$ into elementary molecules (i.e. $\{3\}$, $\{4\}$ and $\{33\}$ molecules, and $\{2\}$ molecules with 2 top or 2 bottom fixed ends), such that

$$\#_{\{33\}} - 10d \cdot \#_{\{4\}} \geq c \cdot \rho, \quad (1.1)$$

where $\#_{\{33\}}$ is the number of $\{33\}$ molecules etc., and c is a positive constant.

The proof of (1.1) for simplified 2-layer models, which exhibits the main thread of the full proof, is further divided into 3 cases, corresponding to **toy models I–III** defined below (for the purpose of presentation, we will consider both toy model I and toy model I plus, which is a natural extension of the former). This classification is based on the following two considerations:

- (1) Consider the set of atoms in $\mathbb{M}_U$ that are connected to *two* atoms in $\mathbb{M}_D$ by two bonds. Is this set big or small relative to ρ ?
- (2) Consider all the atoms in $\mathbb{M}_U$ that are connected to *at least one* atom in $\mathbb{M}_D$ by at least one bond. Are they *all connected* to each other (among themselves, without using the other atoms in $\mathbb{M}_U$)? Or are they *separated* by other atoms in $\mathbb{M}_U$?

The reason for considering these two questions will be clear later. For convenience, throughout this note we will refer to the atoms in (1) (those in $\mathbb{M}_U$ that are connected to *two* atoms in $\mathbb{M}_D$) as *2-connections*, and denote their number by $\#_{2\text{conn}}$. We also define the set in (2) (the atoms in $\mathbb{M}_U$ that are connected to *some* atom in $\mathbb{M}_D$) by X (note that $|X| \gtrsim \rho$), and define the number of components of X (as defined in (2) above) by $\#_{\text{comp}}(X)$.

The four examples in Figure 1, which are examples of toy models I–III below, exactly correspond to different cases of two-layer molecules depending on different answers to the above questions. More precisely, we define a two-layer molecule to be:

- *Toy model I*: if $\#_{2\text{conn}} = 0$ and $X = \mathbb{M}_U$;
- *Toy model I plus*: if $\#_{2\text{conn}} = 0$ and $\#_{\text{comp}}(X) < (100d)^{-1} \cdot |X|$ (no 2-connections, X has few components);
- *Toy model II*: if $\#_{2\text{conn}} = 0$ and $\#_{\text{comp}}(X) \geq (100d)^{-1} \cdot |X|$ (no 2-connections, X has many components);
- *Toy model III*: if $\#_{2\text{conn}} \geq (100d)^{-1} \cdot |X|$ (many 2-connections).

We split the first toy model case into Toy model I and I plus just to simplify the exposition. Note also that, the case when $0 < \#_{2\text{conn}} < (100d)^{-1} \cdot |X|$ is treated similarly to toy models I–II but with an extra step of removing 2-connections, see Section 1.5.5. As such, in later subsections, we will focus on how to treat each toy model by itself.

The plan for this section is as follows. In Section 1.1 we consider the toy model I, whose proof (Proposition 1.6) involves the core arguments in our algorithm in its simplest form. In Section 1.2 we consider the toy model I plus, which is an extension of the toy model I, and can be treated in similar manners. However, here we need to introduce the idea of **UP** algorithm, which will play an important role in the later proofs. In Section 1.3 we consider the toy model II; this requires a different argument, which involves making a clever choice in the **UP** algorithm introduced above. In Section 1.4 we consider the toy model III, where we relies on a simpler argument, which again involves making suitable choices in the **UP** algorithm. Finally, in Section 1.5 we discuss the simplifications made in Definition 1.1 compared to the full generality case, as well as the additions needed to treat the full generality setting.

In Table 1 below we list the different toy models, their defining properties, and the places in the note where they occur. In Table 2 we list the simplifications made in Definition 1.1 compared to the full generality case, as well as the additions needed to treat the full generality setting, and the places in the note where these additions are discussed.

	Case	Property	Location	Algorithm name
(1)	Toy model I	No 2-connections and $X = \mathbb{M}_U$	Section 1.1	(Special case)
(2)	Toy model I plus	No 2-connections X has few components	Section 1.2	MAINUD
(3)	Toy model II	No 2-connections X has many components	Section 1.3	3COMPUP
(4)	Toy model III	Many 2-connections	Section 1.4	2CONNUP/DN

Table 1: The different toy models and their definitions, the places where they occur and the corresponding names of algorithms.

	Simplification	Addition	Discussion
(1)	Ignore O-atoms	General cutting, ov-segments etc.	Section 1.5.1
(2)	Ignore degeneracies	Strong and weak degeneracies	Section 1.5.4
(3)	Each layer is a tree/forest	Layer refining	Section 1.5.3
(4)	Have only two layers	Layer selection	Section 1.5.2

Table 2: Simplifications in Definition 1.1 and corresponding additions which will be needed in real problem. See Section 1.5 for more details.

We remark that this note is presented with 24 figures to help explain the concepts and constructions. We opted to place those figures at the end of the notes in order not to disturb the continuity of the text.

1.1 Toy model I

In this subsection we consider the following (first) toy model:

Definition 1.2. We say $\mathbb{M}$ is a *toy model I*, if it is a simplified 2-layer model as in Definition 1.1, and that each atom in $\mathbb{M}_U$ has *exactly* one bond connected to an atom in $\mathbb{M}_D$ (i.e. $\#_{2\text{conn}} = 0$ and $X = \mathbb{M}_U$).

Definition 1.3. Let $\mathbb{M}_D$ be part of the simplified 2-layer model as in Definition 1.1 (or any resulting molecule after a cutting sequence). We define $\mathbb{M}_D$ to be *proper*, if it does not contain deg 1 or 2 atoms, does not contain two adjacent deg 3 atoms, and does not contain two deg 3 atoms which are both adjacent to the same deg 4 atom, see Figure 2.

If $\mathbb{M}$ is a toy model I, we define the following algorithm applied to it:

Definition 1.4 (algorithm for toy model I). Let $\mathbb{M}$ be a toy model I. We define the following cutting sequence. For an example see Figure 3:

- (1) If not all atom in $\mathbb{M}_U$ have been cut, then choose a lowest atom $\mathbf{n} \in \mathbb{M}_U$ that has not been cut (i.e. either $\mathbf{n}$ has not child in $\mathbb{M}_U$ or all its children in $\mathbb{M}_U$ have been cut).
- (2) If $\mathbf{n}$ is adjacent to an atom $\mathbf{m} \in \mathbb{M}_D$ which has deg 3, then cut $\{\mathbf{n}, \mathbf{m}\}$ as free and cut $\mathbf{m}$ as free from $\{\mathbf{n}, \mathbf{m}\}$ if $\mathbf{n}$ has deg 4; otherwise just cut $\mathbf{n}$ as free.
- (3) If $\mathbb{M}_D$ contains two adjacent deg 3 atoms (say $\mathbf{r}$ and $\mathbf{r}'$), then cut $\{\mathbf{r}, \mathbf{r}'\}$ as free; repeat until $\mathbb{M}_D$ has no adjacent deg 3 atoms.
- (4) If $\mathbb{M}_D$ contains two degree 3 atoms (say $\mathbf{r}$ and $\mathbf{r}''$) both adjacent to the same deg 4 atom (say $\mathbf{r}'$), then cut $\{\mathbf{r}, \mathbf{r}', \mathbf{r}''\}$ as free and cut $\mathbf{r}$ as free from $\{\mathbf{r}, \mathbf{r}', \mathbf{r}''\}$; then go to (3)–(4) and repeat, until $\mathbb{M}_D$ becomes proper (Definition 1.3).
- (5) Go to (1) and choose the next lowest atom $\mathbf{n} \in \mathbb{M}_U$ that has not been cut, and so on.
- (6) Suppose all atoms in $\mathbb{M}_U$ have been cut. If $\mathbb{M}_D$ is proper, then cut any deg 3 atom; if not, then go to (3)–(4) and repeat to remove them, and next cut any deg 3 atom, and so on.

Remark 1.5. In Definition 1.4, we are essentially applying a “naive” algorithm in $\mathbb{M}_U$ where we simply cut a lowest atom each time, and a “greedy” algorithm in $\mathbb{M}_D$ where we exploit a {33} molecule (or a so-called {343} molecule¹ where two deg 3 atoms are adjacent to the same deg 4 atom). As mentioned above, this is the core idea of our proof, in its bare simplest form. It is a simple special case of the **MAINUD** algorithm, which corresponds to the toy model I plus in Section 1.2, see Definition 1.12.

¹The reason to include this {343} molecule is technical: removing only {33} cannot exclude all bad scenarios, i.e. Lemma 1.7 would not be true if we did this.

Proposition 1.6. Let $\mathbb{M}$ be a toy model I, and we apply the algorithm defined in Definition 1.4. Then in this process we only obtain elementary molecules, and we have

$$\#_{\{4\}} = 1, \quad \#_{\{33\}} \geq (|\mathbb{M}_U| - 1)/5. \quad (1.2)$$

Since $|\mathbb{M}_U| = |X| \gtrsim \rho \gg 1$, this clearly implies (1.1).

Proof. Recall the steps (1)–(6) in Definition 1.4. We divide the proof into 4 parts.

Proof part 1. We first prove that each atom in $\mathbb{M}_U$ is contained in an elementary component (after it is cut). By the choices we make in (1) and (5), we know that for each $\mathbf{n} \in \mathbb{M}_U$, at the time this $\mathbf{n}$ is cut, it must be a *lowest* atom that still remains in $\mathbb{M}_U$; thus no *child* of it still remains in $\mathbb{M}_U$, hence cutting $\mathbf{n}$ will not introduce any *top fixed end* in $\mathbb{M}_U$ (and neither does cutting anything in $\mathbb{M}_D$). This means that $\mathbb{M}_U$ will not contain any top fixed end throughout the process. Since each $\mathbf{n} \in \mathbb{M}_U$ either forms a component by itself (when we cut $\{\mathbf{n}\}$ as free in (2)) or belongs to a $\{33\}$ molecule (when we cut $\{\mathbf{n}, \mathbf{m}\}$ as free in (2)), it is clear by definition of elementary components that $\mathbf{n}$ always belongs to an elementary component.

Proof part 2. We next prove that each atom in $\mathbb{M}_D$ is contained in an elementary component. Note that $\mathbb{M}_D$ initially is a tree, and will remain a forest in the whole cutting process (cutting an atom will not create any cycle, but may cause a tree to become a disconnected forest). Now consider each step in (2) involving $\mathbf{n} \in \mathbb{M}_U$, at which time $\mathbb{M}_D$ must be proper by the construction of our algorithm. Consider the possible effect of this step on $\mathbb{M}_D$ (we may assume that $\mathbf{n}$ is connected to one atom $\mathbf{m} \in \mathbb{M}_D$ by a bond, otherwise cutting $\mathbf{n}$ will have no effect on $\mathbb{M}_D$), see Figure 4:

- (a) If we cut $\mathbf{n}$ as free (which means $\mathbf{m}$ has deg 4), then when viewed in $\mathbb{M}_D$, this step simply turns one and only one free end at $\mathbf{m} \in \mathbb{M}_D$ into a fixed end, turning $\mathbf{m}$ from deg 4 to deg 3.
- (b) If we cut $\{\mathbf{n}, \mathbf{m}\}$ as free (which means $\mathbf{m}$ has deg 3), then when viewed in $\mathbb{M}_D$, this step simply cuts $\mathbf{m}$, which has deg 3, as free from $\mathbb{M}_D$.

Here in (a), we note that $\mathbf{m}$ must have deg 4 before cutting $\mathbf{n}$ (otherwise $\mathbf{m}$ must have deg 3 because $\mathbb{M}_D$ is proper, in which case we will cut $\{\mathbf{n}, \mathbf{m}\}$ by (2)); also when viewed in $\mathbb{M}_D$, the bond between $\mathbf{n}$ and $\mathbf{m}$ is initially viewed as a free end at $\mathbf{m}$, and is then turned into a fixed end after cutting $\mathbf{n}$ as free, turning $\mathbf{m}$ from deg 4 to deg 3. Likewise $\mathbf{m}$ must have deg 3 in (b), and when viewed in $\mathbb{M}_D$, cutting $\{\mathbf{n}, \mathbf{m}\}$ as free is equivalent to cutting $\{\mathbf{m}\}$ as free because $\mathbf{n}$ has no other bonds connecting to other atoms in $\mathbb{M}_D$.

Now, regarding the effects described in (a)–(b), we have the following lemma:

Lemma 1.7. Suppose $\mathbb{M}_D$ is proper at some point of the algorithm, and becomes non-proper after either (a) or (b). Then, $\mathbb{M}_D$ will once more become proper after applying (3)–(4), via cuttings of $\{33\}$ and $\{343\}$ molecules only (with each $\{343\}$ molecule further cut into a $\{3\}$ and a $\{33\}$ molecule), and *without seeing any deg 2 atom*. See Figure 5 for an illustration.

Proof. Note that $\mathbb{M}_D$ is always a forest. We then have the following fact, see Figure 6:

- (♥) Suppose we cut $S \subseteq \mathbb{M}_D$ as free, where S is a connected subset of $\mathbb{M}_D$ (for example, when S contains a single atom, or 2 atoms connected by a bond, or 3 atoms connected by 2 bonds). Then, for each atom $\mathbf{q} \in \mathbb{M}_D \setminus S$ that is connected to an atom in S by a (pre-cutting) bond e , the post-cutting molecule $\mathbb{M}_D \setminus S$ has exactly one component $X_{\mathbf{q}} \ni \mathbf{q}$, which is in bijection with $\mathbf{q}$ (and $\mathbf{q}$ is the unique atom in $X_{\mathbf{q}}$ that is connected to S pre-cutting, via the bond e). When viewed in $X_{\mathbf{q}}$, the bond e is viewed as a free end at $\mathbf{q}$ before cutting, and is turned into a fixed end after cutting.

The proof of $(\heartsuit)$ easily follows from the property of forests. In fact, any q_1 and q_2 (with corresponding edges e_1 and e_2) must belong to different components of $\mathbb{M}_D$ after cutting S ; if not, then (pre-cutting) they can be connected by a path involving atoms in $\mathbb{M}_D \setminus S$, but they can also be connected by a path with all intermediate atoms being in S (using the bonds e_1 and e_2 and the connectivity of S by itself), which then produces a cycle and contradicts the forest assumption. As such, for each q we get a unique post-cutting component X_q , such that e is the only pre-cutting bond between X_q and S . Clearly, when viewed in X_q , the effect of cutting S as free is equivalent to turning one and only one free end at q (the one corresponding to e) into a fixed end.

Now, using $(\heartsuit)$, we can prove Lemma 1.7. Note that right before performing (b) at a deg 3 atom m , any atom in $\mathbb{M}_D$ adjacent to m must have deg 4 (because $\mathbb{M}_D$ is proper). Then by $(\heartsuit)$, $\mathbb{M}_D$ will break into at most 2 components after performing (b), and the effect of (b) *when viewed in each component* is equivalent to (a). Therefore, we only need to prove Lemma 1.7 for (a).

Assume now $\mathbb{M}_D$ becomes non-proper, say a $\{33\}$ molecule forms, after (a). Let this component be $\{r, r'\}$, where r' has deg 3 before (a), while r has deg 4 before (a) and is turned into deg 3 after (a). Then, *before (a)*, any atom in $\mathbb{M}_D$ adjacent to r or r' *must have deg 4* (see Figure 7) because otherwise we either have two adjacent deg 3 atoms before (a), or have two deg 3 atoms adjacent to the same deg 4 atom before (a), contradicting properness. Therefore, by $(\heartsuit)$, we see that cutting the $\{33\}$ molecule breaks $\mathbb{M}_D$ into at most 4 components, and the effect of this cutting *when viewed in each component* is again equivalent to (a). We then induct on the size of $\mathbb{M}_D$ and apply induction hypothesis to each of these components, to conclude the proof.

Finally, assume $\mathbb{M}_D$ becomes non-proper, but with no $\{33\}$ molecule forming, after (a). In this case there must be a $\{343\}$ molecule forming, say $\{r, r', r''\}$, where r' and r'' respectively have deg 4 and 3 before (a), and r has deg 4 before (a) and is turned into deg 3 after (a). Then, *before (a)*, any atom adjacent to r' or r'' *must have deg 4* (see Figure 7) for the same reason as above thanks to properness. Moreover, before (a), any atom adjacent to r must also have deg 4, because otherwise we would get a $\{33\}$ molecule after (a), contradicting our assumption. Therefore, by $(\heartsuit)$, we see that cutting the $\{343\}$ molecule breaks $\mathbb{M}_D$ into at most 6 components, and the effect of this cutting *when viewed in each component* is again equivalent to (a). We then conclude the proof by the same induction as above. This completes the proof of Lemma 1.7. $\square$

Now, with Lemma 1.7, we can finish part 2 of the proof, i.e. each atom in $\mathbb{M}_D$ is contained in an elementary component. In fact, those cut in (2) clearly belong to $\{33\}$ molecule. Those cut in (3)–(4) also belong to $\{33\}$ or $\{343\}$ (and subsequently $\{3\}$ or $\{33\}$) components by Lemma 1.7. Finally, those cut in (6) either belong to $\{3\}$ molecule, or are cut in the process of applying (3)–(4) and thus belong to $\{3\}$ or $\{33\}$ molecules by Lemma 1.7.

Proof part 3. With parts 1 and 2 of the proof, we now know that every component is elementary; it suffices to calculate $\#_{\{4\}}$ and $\#_{\{33\}}$. We first show that $\#_{\{4\}} = 1$. Since the first atom cut in (2) must have deg 4, it suffices to show that $\#_{\{4\}} \leq 1$. In fact, for each $n \in \mathbb{M}_U$, at the time n is cut, it must be the *lowest remaining atom* in $\mathbb{M}_U$, which means any possible child of n in $\mathbb{M}_U$ must have been cut before, creating a fixed end at n . Therefore, *if n has any child in $\mathbb{M}_U$ at all* before any cutting, then it cannot have deg 4 when it is cut.

On the other hand, before any cutting, $\mathbb{M}_U$ *can have at most one atom without child in $\mathbb{M}_U$* (see Figure 8); if not, say there exist two such atoms n and n' , then consider a path $(n = n_0, \dots, n_j = n')$ between n and n' within $\mathbb{M}_U$ (since $\mathbb{M}_U$ is a tree). Then n_1 must be parent of n as n has no child in $\mathbb{M}_U$, and then n_2 must be parent of n_1 as n_1 has a child $n \in \mathbb{M}_U$ and another child in $\mathbb{M}_D$ by Definition 1.2, and so on. In the end $n' = n_j$ must be parent of $n_{j-1} \in \mathbb{M}_U$, which contradicts the assumption on n' . This proves $\#_{\{4\}} = 1$.

Proof part 4. Finally we show that $\#_{\{33\}} \geq (|\mathbb{M}_U| - 1)/5$. Consider the behavior of the quantity

$$\lambda := \#_{\text{deg3}} - \#_{\text{comp}},$$

where $\#_{\text{deg3}}$ and $\#_{\text{comp}}$ represent the number of (remaining) deg 3 atoms and components in $\mathbb{M}_D$ respectively, so that initially $\lambda = -1$ and finally $\lambda = 0$. Note that when viewed from $\mathbb{M}_D$, each step in (1)–(6) is one of the followings:

- Operation (a) or (b) defined above (which occur in (2));
- Cutting $\{33\}$ or $\{343\}$ molecule as free (which occur in (3)–(4) and (6));
- Cutting $\{3\}$ molecule as free (which occur in (6)).

Define the number of these operations to be $\#_{(a)}$, $\#_{(b)}$ and $\#_{\text{cut}\{33\}}$ etc., then we have

$$\#_{\{33\}} = \#_{(b)} + \#_{\text{cut}\{33\}} + \#_{\text{cut}\{343\}}. \quad (1.3)$$

Moreover, using $(\heartsuit)$ in the proof of Lemma 1.7 (and the fact that $\mathbb{M}_D$ is proper right before each (a) or (b) or cutting of $\{3\}$), it is easy to see that

- Operation (a) increased λ by 1, and operation (b) does not change λ ;
- Cutting $\{33\}$ or $\{343\}$ decreases λ by 1, and cutting $\{3\}$ does not change λ .

Using the initial and final values of λ we deduce that

$$\#_{(a)} - (\#_{\text{cut}\{33\}} + \#_{\text{cut}\{343\}}) = 1. \quad (1.4)$$

Finally, consider the bonds between $\mathbb{M}_U$ and $\mathbb{M}_D$ (there are $|\mathbb{M}_U|$ of them). Each such bond is broken either in one step (2) (which corresponds to exactly one operation (a) or (b)), or in cutting one of $\{33\}$ or $\{343\}$ molecules; moreover each (a) / each (b) / each cutting of $\{33\}$ / each cutting of $\{343\}$ can break at most 1, 2, 3 and 4 such bonds respectively. This implies that $\#_{(a)} + 2\#_{(b)} + 3\#_{\text{cut}\{33\}} + 4\#_{\text{cut}\{343\}} \geq |\mathbb{M}_U|$. Combining this with (1.3) and (1.4), we deduce that $\#_{\{33\}} \geq (|\mathbb{M}_U| - 1)/5$, as desired. $\square$

1.2 Toy model I plus

In this subsection we extend the toy model I to allow more general connections between $\mathbb{M}_U$ and $\mathbb{M}_D$.

Definition 1.8. We say $\mathbb{M}$ is a *toy model I plus*, if it is a simplified 2-layer model as in Definition 1.1, and each atom $\mathbf{n} \in \mathbb{M}_U$ has *at most* one bond connected to an atom in $\mathbb{M}_D$ (i.e. $\#_{2\text{conn}} = 0$), and the number of connected components of X satisfies

$$\#_{\text{comp}(X)} \leq (100d)^{-1} \cdot |X|, \quad (1.5)$$

where X is defined at the beginning of this note. Here, if two atoms in X can only be connected via atoms in $\mathbb{M}_U \setminus X$, then they belong to different components of X .

Let us explain the intuition behind Definition 1.8, in particular the role of (1.5). In the case of toy model I, we have $X = \mathbb{M}_U$ which is connected, hence $\#_{\text{comp}(X)} = 1$. Moreover, as in part 3 of the proof of Proposition 1.6, we see that $\mathbb{M}_U$ has at most one lowest atom (i.e. atom without child in $\mathbb{M}_U$; in fact it is not hard to see that $\mathbb{M}_U$ must have an “upside down binary tree” shape). This latter fact allows us to apply a “naive” strategy in $\mathbb{M}_U$ where we *always* cut the lowest remaining atom, *without producing too many bad $\{4\}$ atoms*.

In the current case, $\mathbb{M}_U$ may have many lowest atoms, so we cannot always choose and cut a lowest remaining atom. Instead, we improve the strategy by choosing a *lowest remaining deg 3* atom $\mathbf{n}$, and cutting it and all its descendants (which all have deg 4 when $\mathbf{n}$ is chosen). Here we do not consider deg 2 atoms, as they will always be lowest in $\mathbb{M}_U$ and always have to be cut at top priority. This leads to the idea of the **UP** algorithm, see Definition 1.9 below, which is another basic component in our proof.

Note, however, that this **UP** algorithm cannot be simply combined with the “greedy” algorithm in $\mathbb{M}_D$ as in Definition 1.4, because now $\mathbf{n}$ may have deg 4 descendants in $\mathbb{M}_U$, and this naive attempt may break the key monotonicity property (see part 1 of the proof of Proposition 1.13) and lead to non-elementary components. As such, we need to be careful and only apply **UP** at places in $\mathbb{M}_U$ which will not affect $\mathbb{M}_D$, i.e. in $\mathbb{M}_U \setminus X$. Within X we still need to apply the naive strategy, which may lead to bad $\{4\}$ molecules due to X .

We can then see the role of (1.5) and the *dichotomy* concerning $\#_{\text{comp}(X)}$; in fact, same as in part 3 of the proof of Proposition 1.6, *each component* of X can have *at most one lowest atom*. In the naive strategy, this leads to bad $\{4\}$ molecules whose number is comparable to $\#_{\text{comp}(X)}$. Under the assumption (1.5), this is negligible compared to the number of good $\{33\}$ molecules, which will be comparable to $|X|$ (i.e. the number of bonds between $\mathbb{M}_U$ and $\mathbb{M}_D$).

With the above intuition, we start by defining the **UP** algorithm in the toy model case.

Definition 1.9 (The **UP** algorithm, toy model case). Let $\mathbb{M}$ be any molecule which *does not have top fixed end*. Define the following cutting sequence. For an example see Figure 9:

- (1) If $\mathbb{M}$ has a deg 2 atom, then cut it as free. Repeat until $\mathbb{M}$ does not have deg 2 atoms.
- (2) Choose a lowest deg 3 atom $\mathbf{n}$ that has not been cut (or a lowest atom if $\mathbb{M}$ has no deg 3 atoms). Let $S_{\mathbf{n}}$ be the set of descendants of $\mathbf{n}$. This $\mathbf{n}$ and $S_{\mathbf{n}}$ is fixed until all atoms in $S_{\mathbf{n}}$ have been cut.
- (3) Starting from $\mathbf{n}$, each time choose a highest atom $\mathbf{m}$ in $S_{\mathbf{n}}$ that has not been cut. If $\mathbf{m}$ has deg 3 and has a parent $\mathbf{m}^+$ or child $\mathbf{m}^-$ that also has deg 3, then cut $\{\mathbf{m}, \mathbf{m}^\pm\}$ as free; otherwise cut $\mathbf{m}$ as free. Repeat until all atoms in $S_{\mathbf{n}}$ have been cut, then go to (1)–(2) and choose the next $\mathbf{n}$, and so on.

Remark 1.10. The **UP** algorithm is a combination of a “global low-to-high” (where each time we select the *lowest* remaining deg 3 atom in (2)) and “local high-to-low” procedure (where each time we cut the *highest* remaining atom in $S_{\mathbf{n}}$ in (3)). This leads to a key *monotonicity* property, see Proposition 1.11.

The step (3) in the **UP** algorithm may seem complicated, we explain it as follows: let $\mathbf{m}$ be the highest remaining atom in $S_{\mathbf{n}}$, the simple option is to just cut $\mathbf{m}$ as free. However, in the case described in (3), the $\mathbf{m}^\pm$ will become deg 2 once $\mathbf{m}$ is cut as free; moreover $\mathbf{m}^- \in S_{\mathbf{n}}$ will be a highest remaining atom in $S_{\mathbf{n}}$, and $\mathbf{m}^+ \notin S_{\mathbf{n}}$ will occur as a deg 2 atom in (1) once all atoms in $S_{\mathbf{n}}$ have been cut. As such, this deg 2 atom $\mathbf{m}^\pm$ have to be cut at top priority, so instead of cutting $\{\mathbf{m}\}$ we can combine this with the cutting of $\mathbf{m}^\pm$, which also forms a good $\{33\}$ molecule $\{\mathbf{m}, \mathbf{m}^\pm\}$. This then leads to the step (3) in the **UP** algorithm.

We next prove a key monotonicity property of the **UP** algorithm and a sufficient condition for it to produce at least one $\{33\}$ molecule.

Proposition 1.11. The **UP** algorithm preserves the following *monotonicity* property: at any time, any atom in $(\mathbb{M} \setminus S_{\mathbf{n}}) \cup \{\mathbf{n}\}$ has no *top fixed end*, and any atom in $S_{\mathbf{n}} \setminus \{\mathbf{n}\}$ has no *bottom fixed end*. In particular, applying **UP** only produces elementary components, and with $\#_{\{4\}} \leq 1$ if $\mathbb{M}$ is connected.

Moreover, assume $\mathbb{M}$ is connected and has no deg 2 atom. Then applying **UP** produces only $\{4\}$ and $\{3\}$ molecules, *if and only if* $\mathbb{M}$ is a tree and has at most one deg 3 atom. Otherwise, applying **UP** will produce at least one $\{33\}$ molecule.

Proof. We divide the proof into 4 parts.

Proof part 1. We first prove the monotonicity property, which implies all components are elementary (since each atom either forms a component by itself or belongs to a $\{33\}$ molecule, due to (1)–(3) in Definition 1.9). Note that initially there is no S_n and $\mathbb{M}$ has no top fixed end by assumption, so it suffices to prove that the monotonicity property is perserved. First, any deg 2 atom n cut in (1) must have two *bottom* fixed ends by monotonicity, so it has no child remaining when it is cut, thus cutting it will not introduce any *top fixed end*.

Now let n and S_n be as in (2), and m be a highest remaining atom in S_n as in (3). When n is chosen in (2), all atoms in $S_n \setminus \{n\}$ *must have deg 4* by construction, and thus have no bottom fixed end. Consider the operation of cutting m as free in (3). We then know that that this cutting does not create any *bottom fixed end* in $S_n \setminus \{n\}$ (as m is highest), and does not create any *top fixed end* in $(\mathbb{M} \setminus S_n) \cup \{n\}$ (as any child of m must belong to $S_n \setminus \{n\}$), see Figure 10. The same holds for the operation of cutting $\{m, m^\pm\}$, because this is equivalent to cutting $\{m\}$ and then cutting a deg 2 atom (either $m^+ \notin S_n$ with two bottom fixed ends or $m^- \in S_n \setminus \{n\}$ with two top fixed ends). This proves that the monotonicity property is perserved, as desired.

Proof part 2. We prove $\#_{\{4\}} \leq 1$ when $\mathbb{M}$ is connected. For this, just note that after the first cutting, $\mathbb{M}$ will *always* contain an atom that is not deg 4. By construction, this means that the n chosen in (2) will always be deg 3; moreover, any $m \in S_n \setminus \{n\}$ that is cut in (3) also cannot be deg 4, because m has a parent in S_n which must have been cut before m thanks to the highest assumption of m . Therefore, *only* the first atom can produce a possible $\{4\}$ molecule, and thus $\#_{\{4\}} \leq 1$.

Proof part 3. Suppose $\mathbb{M}$ is connected and has no deg 2 atom, we prove that either $\#_{\{33\}} \geq 1$, or all components are $\{4\}$ and $\{3\}$ (with $\#_{\{4\}} \leq 1$). If not, then we must have a $\{2\}$ component. Consider the first atom m that has deg 2 when we cut it, and consider the cutting operation that turns m into deg 2, which must be cutting some deg 3 atom p as free (this p cannot have deg 4, because a degree 4 atom is cut only when no degree 3 atoms are present, and in this case cutting a degree 4 atom cannot produce a degree 2 atom). This cutting creates a fixed end at m , so this p must be a parent or child of m ; moreover this m and this p must both be deg 3 at the time of cutting p , but this violates Definition 1.9 (3).

Proof part 4. Finally we prove $\#_{\{33\}} = 0$ if and only if $\mathbb{M}$ is a tree and has at most one deg 3 atom. For this, consider the increment of the quantity $\sigma := \#_{\text{bo/fr}} - 3\#_{\text{atom}}$, where $\#_{\text{bo/fr}}$ is the number of bonds and free ends (not counting fixed ends) and $\#_{\text{atom}}$ is the number of remaining atoms. It is easy to see that cutting each $\{2\} / \{3\} / \{33\} / \{4\}$ molecule increases σ by 1, 0, 1 and -1 respectively. Since finally $\sigma = 0$, we see that $\#_{\{4\}} - (\#_{\{2\}} + \#_{\{33\}}) = \sigma_{\text{init}}$ (initial value of σ).

Note that initially $\#_{\text{bo/fr}} = \#_{\text{deg}} - \#_{\text{bond}}$ where $\#_{\text{deg}}$ is the total degree of all atoms (not counting fixed ends) and $\#_{\text{bond}}$ is the number of bonds (because each bond contributes twice to the degrees), so $\sigma_{\text{init}} = (|\mathbb{M}| - \#_{\text{bond}}) - |\mathbb{M}_3|$ where $|\mathbb{M}_3|$ is the number of (initially) deg 3 atoms in $\mathbb{M}$. Moreover by connectedness we have $\#_{\text{bond}} \geq |\mathbb{M}| - 1$ with equality if and only if $\mathbb{M}$ is a tree, and note also $\#_{\{4\}} \in \{0, 1\}$ (and $\#_{\{4\}} = 1$ if and only if $|\mathbb{M}_3| = 0$ by the same proof in part 2), it is then easy to see that $\#_{\{2\}} + \#_{\{33\}} = 0$ if and only if $\mathbb{M}$ is a tree and $|\mathbb{M}_3| \leq 1$. This completes the proof. $\square$

Now we can define the algorithm for toy model I plus. As described in the above heuristics, this features a naive algorithm (always cutting the lowest) in X , an **UP** algorithm in $\mathbb{M}_U \setminus X$, and a greedy algorithm in $\mathbb{M}_D$. Up to some simplifications, this is just the **MAINUD** algorithm, which is the main algorithm in the 2-layer proof. Note also that the part of Definition 1.12 involving $\mathbb{M}_D$ is exactly the same as Definition 1.4; the only difference occurs in $\mathbb{M}_U$ (in fact in $\mathbb{M}_U \setminus X$).

Definition 1.12 (Algorithm for toy model I plus). Let $\mathbb{M}$ be a toy model I plus with X be defined in the beginning of this note (this X is fixed throughout the algorithm). We define the following cutting sequence. For an example see Figures 11–12:

- (1) If $\mathbb{M}_U$ has a deg 2 atom, then cut it as free. Repeat until $\mathbb{M}_U$ does not have deg 2 atoms.
- (2) If not all atom in $\mathbb{M}_U$ have been cut, then choose an atom $\mathbf{n}$ which is a *lowest remaining atom in $\mathbb{M}_U$ that either has deg 3 or belongs to X* , i.e. $\mathbf{n} \in \mathbb{M}_U$ either has deg 3 or belongs to X , and *no other descendant of $\mathbf{n}$ in $\mathbb{M}_U$ has the same property*. Let $S_{\mathbf{n}}$ be the set of descendants of $\mathbf{n}$ in $\mathbb{M}_U$. This $\mathbf{n}$ and $S_{\mathbf{n}}$ is fixed until the end of (4).
- (3) Starting from $\mathbf{n}$, each time choose a highest atom $\mathbf{p}$ in $S_{\mathbf{n}}$ that has not been cut. If $\mathbf{p}$ is adjacent to an atom $\mathbf{q} \in \mathbb{M}_D$ which has deg 3, then cut $\{\mathbf{p}, \mathbf{q}\}$ as free and cut $\mathbf{q}$ as free from $\{\mathbf{p}, \mathbf{q}\}$ if $\mathbf{p}$ has deg 4; otherwise just cut $\mathbf{p}$ as free.
- (4) If $\mathbb{M}_D$ becomes non-proper, then perform the same steps (3)–(4) in Definition 1.4 until it becomes proper again. Then go to (3) and repeat, until all atoms in $S_{\mathbf{n}}$ have been cut.
- (5) When all atoms in $S_{\mathbf{n}}$ have been cut, go to (1)–(2) and choose the next $\mathbf{n}$, and so on.
- (6) When all atoms in $\mathbb{M}_U$ have been cut, then perform the same step (6) in Definition 1.4 to finish off cutting $\mathbb{M}_D$.

Proposition 1.13. Let $\mathbb{M}$ be a toy model I plus, and we apply the algorithm defined in Definition 1.12. Then in this process we only obtain elementary molecules, and we have

$$\#\{4\} \leq \#\text{comp}(X), \quad \#\{33\} \geq (|X| - 1)/5 - \#\text{comp}(X). \quad (1.6)$$

Since $|X| \gtrsim \rho \gg 1$ and using (1.5), this clearly implies (1.1).

Proof. We divide the proof into 4 parts just as Proposition 1.6.

Proof part 1. We show that each atom in $\mathbb{M}_U$ is contained in an elementary component. As in part 1 of the proof of Proposition 1.11, this follows from the same monotonicity property but *for $\mathbb{M}_U$ only*: any atom in $(\mathbb{M}_U \setminus S_{\mathbf{n}}) \cup \{\mathbf{n}\}$ has no *top fixed end*, and any atom in $S_{\mathbf{n}} \setminus \{\mathbf{n}\}$ has no *bottom fixed end*.

To prove the monotonicity property, we first note that, for any $\mathbf{n}$ chosen in (2) and at the time it is chosen, each atom in $S_{\mathbf{n}} \setminus \{\mathbf{n}\}$ has deg 4 and thus has no bottom fixed end. Then consider the process of cutting in (3) and below, see Figure 13. Following exactly part 1 of the proof of Proposition 1.11, we see that the cutting of any atom in $\mathbb{M}_U$ in this process does not create any top fixed end in $(\mathbb{M}_U \setminus S_{\mathbf{n}}) \cup \{\mathbf{n}\}$ or any bottom fixed end in $S_{\mathbf{n}} \setminus \{\mathbf{n}\}$; moreover, for the cutting of any atom in $\mathbb{M}_D$ in this process (such as the one in (4)), it *only creates bottom fixed ends* in $\mathbb{M}_U$, and *only at atoms in $(\mathbb{M}_U \setminus S_{\mathbf{n}}) \cup \{\mathbf{n}\}$* . This is because $S_{\mathbf{n}} \setminus \{\mathbf{n}\}$ contains no atom in X by construction, and thus has no bond connected to $\mathbb{M}_D$, hence no fixed end can be created at any atom in this set when cutting anything in $\mathbb{M}_D$. This proves the monotonicity property.

Proof part 2. We show that each atom in $\mathbb{M}_D$ is contained in an elementary component. Since the part of Definition 1.12 involving $\mathbb{M}_D$ is exactly the same as Definition 1.4, this part is also exactly the same as part 2 of the proof of Proposition 1.6.

Proof part 3. We show that $\#\{4\} \leq \#\text{comp}(X)$. Same as part 2 of the proof of Proposition 1.11, we see that any $\mathbf{p} \in S_{\mathbf{n}} \setminus \{\mathbf{n}\}$ in (3) cannot have deg 4 when it is cut. Therefore, the only possible $\{4\}$ atoms must come from those $\mathbf{n}$ chosen in (2). By our choice in (2), such $\mathbf{n}$ must belong to X and must be *lowest atoms* in X , i.e. it belongs to X but does not have any child in X before any cutting.

Now we claim that any two lowest atoms in X *must belong to different components of X* (see Figure 8). If not, say $\mathbf{n}$ and $\mathbf{n}'$ are connected by a path $(\mathbf{n} = \mathbf{n}_0, \dots, \mathbf{n}_j = \mathbf{n}')$ in X , then arguing as part 3 of the proof of Proposition 1.6 and noticing that each atom in X has a child in $\mathbb{M}_D$, we deduce that each $\mathbf{n}_i$ must be parent of $\mathbf{n}_{i-1}$ which then leads to contradiction with properties of $\mathbf{n}_j = \mathbf{n}'$. We then conclude that the number of lowest atoms in X does not exceed $\#\text{comp}(X)$, hence $\#\{4\} \leq \#\text{comp}(X)$.

Proof part 4. Finally we show that $\#_{\{33\}} \geq (|X| - 1)/5 - \#_{\text{comp}(X)}$. Since the part of Definition 1.12 involving $\mathbb{M}_D$ is exactly the same as Definition 1.4, this part is essentially the same as part 4 of the proof of Proposition 1.6. There is only one difference: in the current case, each step of cutting $\{\mathbf{p}, \mathbf{q}\}$ as free in (3) (which corresponds to the operation (b) in part 4 of the proof of Proposition 1.6 when viewed in $\mathbb{M}_D$) may produce two $\{3\}$ molecules instead of one $\{33\}$ molecule, but this only happens when² $\mathbf{p}$ has deg 4, and the number of such exceptions is bounded by $\#_{\text{comp}(X)}$ following part 3 above. Therefore, instead of (1.3) we have

$$\#_{\{33\}} \geq \#_{(b)} + \#_{\text{cut}\{33\}} + \#_{\text{cut}\{343\}} - \#_{\text{comp}(X)}, \quad (1.7)$$

and the same proof then yields that, instead of $\#_{\{33\}} \geq (|X| - 1)/5$ as in (1.2), we now have $\#_{\{33\}} \geq (|X| - 1)/5 - \#_{\text{comp}(X)}$. This completes the proof. $\square$

1.3 Toy model II

In this subsection we introduce the toy model II, which is opposite to toy model I plus via the dichotomy, where we assume $\#_{\text{comp}(X)} \gtrsim |X|$ instead of $\#_{\text{comp}(X)} \ll |X|$ in Definition 1.8.

Definition 1.14. We say $\mathbb{M}$ is a toy model II, if it satisfies all the assumptions in Definition 1.8 for toy model I plus with the same notations, except that (1.5) is replaced by $|X| \geq 2$ and

$$\#_{\text{comp}(X)} > (100d)^{-1} \cdot |X|. \quad (1.8)$$

Let us explain the strategy for toy model II, which will be completely different from toy models I and I plus. Here we will not gain from the greedy algorithm in $\mathbb{M}_D$; instead we first cut all of $\mathbb{M}_D$ as free without exploiting any gain, and then gain from $\mathbb{M}_U$ under the assumption (1.8). Note that, after cutting $\mathbb{M}_D$ as free, there will not be any deg 2 atoms in $\mathbb{M}_U$, and the deg 3 atoms in $\mathbb{M}_U$ are *exactly those in X* .

The intuition is that, due to the lower bound of the number of components if X given by (1.8), the deg 3 atoms in $\mathbb{M}_U$ are now *separated by deg 4 atoms*. Essentially, we expect that for such molecules, we can design a cutting algorithm with minimal loss caused $\{4\}$ molecules, such that *each connected component of the set of deg 3 atoms* provides a good $\{33\}$ molecule. In fact, this can be achieved by a clever choice in the **UP** algorithm. We next introduce the main algorithm in this case (together with an auxiliary lemma needed in this algorithm). Up to simplifications, this is just the **3COMPUP** algorithm.

Lemma 1.15. Suppose $\mathbb{M}$ is a molecule that is a tree, contains only deg 3 and 4 atoms and has no top fixed end. Let the set of deg 3 atoms in $\mathbb{M}$ be $\mathbb{M}_3$ with $|\mathbb{M}_3| \geq 2$. Define a set $A \subseteq \mathbb{M}$ to be *no-bottom* if any child of any atom in A must also be in A (see Figure 14). Then, there always exists a connected, no-bottom subset $A \subseteq \mathbb{M}$ with $|A \cap \mathbb{M}_3| = 2$.

Proof. For any $\mathbf{n} \in \mathbb{M}$, let $S_{\mathbf{n}}$ be the set of descendants of $\mathbf{n}$ (including $\mathbf{n}$ itself), then it is always connected and no-bottom. If there exists $\mathbf{n}$ such that $|S_{\mathbf{n}} \cap \mathbb{M}_3| \geq 2$, then choose a lowest such $\mathbf{n}$. Since $\mathbf{n}$ has no top fixed end, we know that either (i) $\mathbf{n}$ has one child $\mathbf{n}_1$, or (ii) $\mathbf{n}$ has deg 4 and has two children $\mathbf{n}_1, \mathbf{n}_2$. In either case, by assumption we have $|S_{\mathbf{n}_j} \cap \mathbb{M}_3| \leq 1$ for each j , which then implies $|S_{\mathbf{n}} \cap \mathbb{M}_3| \leq 2$ because $S_{\mathbf{n}} = (\cup_j S_{\mathbf{n}_j}) \cup \{\mathbf{n}\}$. This implies $|S_{\mathbf{n}} \cap \mathbb{M}_3| = 2$, as desired.

Now assume $|S_{\mathbf{n}} \cap \mathbb{M}_3| \leq 1$ for each $\mathbf{n} \in \mathbb{M}$. If $|S_{\mathbf{n}} \cap \mathbb{M}_3| = 1$, there is a unique $\mathbf{p} = \mathbf{p}(\mathbf{n}) \in S_{\mathbf{n}} \cap \mathbb{M}_3$; now consider all atom pairs $(\mathbf{n}, \mathbf{n}')$ such that $\mathbf{p}(\mathbf{n})$ and $\mathbf{p}(\mathbf{n}')$ both exist and *are different* (which exist because $|\mathbb{M}_3| \geq 2$), and one such pair $(\mathbf{n}, \mathbf{n}')$ that are connected by a *shortest path* in $\mathbb{M}$. Let this path be

²This cannot happen for toy model I, because there $\mathbf{p}$ can have deg 4 only when it is the first atom cut, in which case $\mathbf{q}$ cannot have deg 3.

$(n_0, n, \dots, n_j = n')$, then for each $0 < i < j$ we must have either $S_{n_i} \cap \mathbb{M}_3 = \emptyset$ or $p(n_i) = p(n)$ (otherwise the pair (n, n_i) corresponds to a shorter path, contradiction). Now let $A = \cup_{i=0}^j S_{n_i}$, it is clear that A is connected, no-bottom and $A \cap \mathbb{M}_3 = \{p(n), p(n')\}$, as desired. $\square$

Definition 1.16 (The **3COMPUP** algorithm, toy model case). Let $\mathbb{M}$ be a molecule that is a tree, contains only deg 3 and 4 atoms and has no top fixed end. Let the set of deg 3 atoms in $\mathbb{M}$ be $\mathbb{M}_3$ with $|\mathbb{M}_3| \geq 2$. Define the following cutting sequence. For an example see [Figure 15](#):

- (1) Apply Lemma 1.15 to $\mathbb{M}$ and select a connected no-bottom subset A . This set is fixed until the end of (5).
- (2) Choose a lowest deg 3 atom n in A that has not been cut. Let S_n be the set of descendants of n . This n and S_n are fixed until the end of (3).
- (3) Starting from n , each time choose a highest atom m in S_n that has not been cut. If m has deg 3 and has a parent m^+ or child m^- that also has deg 3, then cut $\{m, m^\pm\}$ as free; otherwise cut m as free. Repeat until all atoms in S_n have been cut.
- (4) If $\mathbb{M}$ has deg 2 atom, then cut it; repeat until $\mathbb{M}$ has no deg 2 atoms. Then go to (2) and choose the next n , and so on.
- (5) When all atoms in A have been cut, we subsequently cut all deg 2 atoms as in (4), and then cut all *components of $\mathbb{M}$ with only one deg 3 atom* using **UP**.
- (6) Go to (1) and select the next A from one of the remaining components of $\mathbb{M}$, and so on.

Remark 1.17. Let us explain the ideas in Lemma 1.15 and Definition 1.16. Note that steps (2)–(4) are exactly the same as **UP** but applied to a particular set A , so we can view **3COMPUP** as a clever way of carrying out **UP**. The role of the set A is explained as follows: first it needs to be connected (otherwise we can cut each of its components separately); it should also be no-bottom, because when cutting any atom we should also cut its descendants to aguarantee the monotonicity property. Finally we need it to contain two deg 3 atoms so that this sequence of cutting will produce at least one $\{33\}$ molecule, and we need *exactly two* deg 3 atoms so that we do not mess up with too many components of deg 3 atoms (which are the key objects leading to the gain).

Proposition 1.18. Let $\mathbb{M}$ be a toy model II. We first cut $\mathbb{M}_D$ as free and cut it into elementary components by **DOWN** (i.e. the dual of **UP** with all directions reversed), then apply **3COMPUP** (Definition 1.16) to $\mathbb{M}_U$. Then all resulting components are elementary, and we have

$$\#_{\{4\}} = 1, \quad \#_{\{33\}} \geq \#_{\text{comp}(X)}/10. \quad (1.9)$$

Since $|X| \gtrsim \rho \gg 1$ and using (1.8), this clearly implies (1.1).

Proof. Note that steps (2)–(4) in Definition 1.16 are exactly the same as **UP**, so the proof that all components are elementary (as well as $\#_{\{4\}} = 1$ which comes only from $\mathbb{M}_D$) follows in the same way as Proposition 1.11. We now focus on the proof that $\#_{\{33\}} \geq \#_{\text{comp}(X)}/10$.

For any molecule $\mathbb{M}$, define $\#_{\text{comp}(\mathbb{M})}$ to be the number of components of $\mathbb{M}$, and $\#_{3\text{comp}(\mathbb{M})}$ to be the number of components of the set of deg 3 atoms in $\mathbb{M}$, and

$$\lambda(\mathbb{M}) := \frac{1}{5} \#_{\text{comp}(\mathbb{M})} + \frac{1}{10} \#_{3\text{comp}(\mathbb{M})}. \quad (1.10)$$

In Definition 1.16 (applied to $\mathbb{M}_U$), the whole algorithm consists of finitely many loops where one loop is formed by steps (1)–(5). Consider each such loop which reduces $\mathbb{M}$ to $\mathbb{M}'$, we shall prove the following:

(♣) If $\mathbb{M}$ is a tree, has no top fixed end and contains only deg 3 and 4 atoms with at least two deg 3 atoms, then each component of $\mathbb{M}'$ satisfies the same property. Moreover for this loop we have

$$\#_{\{33\}} \geq \lambda(\mathbb{M}) - \lambda(\mathbb{M}'). \quad (1.11)$$

Note that adding up (1.11) leads to the desired lower bound of $\#_{\{33\}}$ (initially $\mathbb{M}_U$ has no deg 2 atoms nor top fixed end, and $\#_{3\text{comp}}(\mathbb{M}_U) = \#_{\text{comp}}(X)$), thus it suffices to prove (♣) for a given loop. We divide the proof into 4 parts.

Proof part 1. We prove that each component of $\mathbb{M}'$ satisfies the desired properties. This is obvious, as the lack of top fixed ends follows from the same proof as Proposition 1.11, and the lack of deg 2 atoms and components with only one deg 3 atom follows from the cuttings done in (5).

Proof part 2. Now we need to prove (1.11). We start by making some preparations. Note that A is a connected subset of $\mathbb{M}$; by (♥) in the proof of Lemma 1.7, each atom $\mathbf{m} \in \mathbb{M} \setminus A$ that is connected to an atom $\mathbf{n} = \mathbf{n}(\mathbf{m})$ in A by a pre-loop bond e corresponds to a unique component $X_{\mathbf{m}}$ after cutting the atoms in A as free, see Figure 16 (the loop will in fact cut as free more atoms). Note also that $\mathbf{m}$ must be a parent of $\mathbf{n}$ due to the no-bottom property of A .

Let $\mathbb{M}_3$ be the set of deg 3 atoms in $\mathbb{M}$ (before executing the loop), and consider all components of $\mathbb{M}_3$ that intersect A or are connected to A by a bond. We know A has exactly two atoms in $\mathbb{M}_3$; apart from the one or two components that contain these two atoms, each component under consideration must be a subset of a unique $X_{\mathbf{m}}$ where $\mathbf{m}$ has deg 3 and $\mathbf{n} = \mathbf{n}(\mathbf{m})$ has deg 4 (call it $Y_{\mathbf{m}}$). Let B be the union of these components with A (Figure 16). We claim that after steps (1)–(4) and cutting the deg 2 atoms in (5), exactly those atoms in B have been cut as free.

In fact, the atoms A must have been cut by definition. For any atom in the $\mathbb{M}_3$ components forming B , by definition it is connected to an atom in A by a chain of deg 3 atoms; once the atom in A has been cut, the atoms on this chain will successively become deg 2, so they must have been cut (either in (4)–(5), or as part of $\{33\}$ molecule in (3)). On the other hand, each atom in $\mathbb{M} \setminus B$ has at most one bond connected to B (as B is connected and $\mathbb{M}$ is a tree), and must be deg 4 if it has one such bond (otherwise it would belong to one of the $\mathbb{M}_3$ components forming B), so each atom in $\mathbb{M} \setminus B$ will have deg 3 or 4 after cutting B as free, thus it will not be cut in steps (1)–(4) or in cutting the deg 2 atoms in (5).

Proof part 3. Next we obtain a lower bound on $\#_{\{33\}}$. Note that A contains two deg 3 atoms, we can show that $\#_{\{33\}} \geq 1$ in the same way as in part 3 of the proof of Proposition 1.11. Moreover, since A contains *exactly* two deg 3 atoms, by arguing as in part 4 of the proof of Proposition 1.11 and calculating the quantity σ defined there but within A , we can show that, apart from at most two exceptions, each deg 4 atom in A will have deg 3 when it is cut.

Now let λ be the number of $\mathbb{M}_3$ components forming B . With at most two exceptions, each such component corresponds to a deg 3 atom $\mathbf{m} \in X_{\mathbf{m}}$ together with its deg 4 child $\mathbf{n} = \mathbf{n}(\mathbf{m}) \in A$ as in part 2 above. With at most two more exceptions, this $\mathbf{n}$ will have deg 3 when it is cut, at which time $\mathbf{m}$ also has deg 3 (see Figure 17); then by (3), we know that $\mathbf{n}$ must belong to a $\{33\}$ molecule (which may or may not be $\{\mathbf{n}, \mathbf{m}\}$). Note that each $\{33\}$ molecule can be so obtained by at most one $\mathbf{n}$, and each $\mathbf{n}$ can be obtained by at most two $\mathbf{m}$ (hence at most two components), we conclude that

$$\#_{\{33\}} \geq \max \left(1, \frac{\lambda - 2}{2} - 2 \right). \quad (1.12)$$

Proof part 4. Now we consider the components of $\mathbb{M}'$ and $\mathbb{M}'_3$ (set of deg 3 atoms in $\mathbb{M}'$). Note that $\mathbb{M}'$ is formed from $\mathbb{M}$ by cutting B as free and then removing all components with only one deg 3 atom. Since B is connected, by applying (♥) in the proof of Lemma 1.7 again, we see that each component $X_{\mathbf{q}}$ of $\mathbb{M}'$ corresponds to a unique atom $\mathbf{q} \in \mathbb{M} \setminus B$ that is connected to an atom in B , see Figure 18. Let μ

be the number of such bonds (i.e. corresponding to those new components with at least two deg 3 atoms after cutting B), then we have

$$\#_{\text{comp}}(\mathbb{M}) - \#_{\text{comp}}(\mathbb{M}') = 1 - \mu. \quad (1.13)$$

Finally, consider the set of deg 3 atoms, i.e. $\mathbb{M}_3$ and $\mathbb{M}'_3$, and their components. The components of $\mathbb{M}_3$ includes those λ components forming B as in part 2 (which have been cut when forming $\mathbb{M}'$), and the other components which are contained in some X_q defined above. Note that for each new component Y formed after cutting B that contains only one deg 3 atom, it will be removed and will not occur in $\mathbb{M}'$; however, all atoms in Y originally have deg 4 before any cutting, so initially Y also has no contribution to $\#_{3\text{comp}}(\mathbb{M})$, and thus we can ignore such Y .

By $(\heartsuit)$, when viewed in each X_q , the effect of cutting B exactly turns one free end into fixed end (turning one deg 4 atom into deg 3). Therefore the part of $\mathbb{M}'_3$ within X_q is formed by *adding one deg 3 atom* into the existing part of $\mathbb{M}_3$ within X_q (see Figure 18). This new deg 3 atom may *decrease* $\#_{3\text{comp}}(X_q)$ (i.e. the number of components of the set of deg 3 atoms in X_q) by joining existing components together, but this decrease is *at most 2* because this new deg 3 atom can join at most 3 components into 1. From this, and accounting for the λ components removed when cutting B , we deduce that

$$\#_{3\text{comp}}(\mathbb{M}) - \#_{3\text{comp}}(\mathbb{M}') \leq \lambda + 2\mu. \quad (1.14)$$

Putting together (1.10) and (1.12)–(1.14), we deduce that

$$\begin{aligned} \lambda(\mathbb{M}) - \lambda(\mathbb{M}') &= \frac{1}{5}(\#_{\text{comp}}(\mathbb{M}) - \#_{\text{comp}}(\mathbb{M}')) + \frac{1}{10}(\#_{3\text{comp}}(\mathbb{M}) - \#_{3\text{comp}}(\mathbb{M}')) \\ &\leq \frac{1}{5} + \frac{\lambda}{10} \leq \max\left(1, \frac{\lambda - 6}{2}\right) \leq \#\{33\}. \end{aligned} \quad (1.15)$$

This proves (1.11) and completes the proof of Proposition 1.18. $\square$

1.4 Toy model III

In this subsection we introduce the toy model III. Note that in both toy models I plus and II, we are assuming the absence of 2-connections, i.e. $\#_{2\text{conn}} = 0$. Now we define the toy models III, which are (essentially) opposite to toy models I plus and II via the *dichotomy*, where we assume³ $\#_{2\text{conn}} \gtrsim |X|$ instead of $\#_{2\text{conn}} = 0$.

Definition 1.19. We say $\mathbb{M}$ is a toy model III, if it is a simplified 2-layer model in Definition 1.1, and

$$\#_{2\text{conn}} \geq (100d)^{-1} \cdot |X|, \quad (1.16)$$

where $\#_{2\text{conn}}$ and X are defined at the beginning of this note.

The strategy for toy models III is simple: we work in $\mathbb{M}_D$ as if $\mathbb{M}_U$ were absent, and apply the **DOWN** algorithm (the dual of **UP**) to it. Now consider any 2-connection $\mathbf{n} \in \mathbb{M}_U$, which will become deg 2 when all atoms in $\mathbb{M}_D$ have been cut. Assume $\mathbf{n}$ is connected to two atoms $\mathbf{m}_1, \mathbf{m}_2 \in \mathbb{M}_D$, and assume $\mathbf{m}_1$ is cut before $\mathbf{m}_2$ in the **DOWN** algorithm applied to $\mathbb{M}_D$. Then, right before $\mathbf{m}_2$ is cut, this $\mathbf{m}_2$ will have deg 3 (say) by Proposition 1.11; also since $\mathbf{m}_1$ has been cut at this time, now $\mathbf{n}$ must also have deg 3, which leads to a $\{33\}$ molecule, so we simply cut it and enjoy the gain. This way we guarantee that each 2-connection belongs to a $\{33\}$ molecule, which is sufficient under the assumption (1.16). We shall refer to this as the **2CONNNDN** algorithm (and the dual version **2CONNUP**).

³There is also the case $0 < \#_{2\text{conn}} \ll |X|$; in this case we will first cut all such 2-connections, which generates an error which is eventually negligible but still requires some extra care; see Section 1.5.5 for more discussions.

Definition 1.20 (Algorithm for toy model III). Let $\mathbb{M}$ be toy model III. We define the following cutting sequence. For an example see [Figure 19](#):

- (1) If $\mathbb{M}_D$ has a deg 2 atom, then cut it as free. Repeat until $\mathbb{M}_D$ does not have deg 2 atoms.
- (2) Choose a highest deg 3 atom $\mathbf{n}$ in $\mathbb{M}_D$ that has not been cut (or a highest atom if $\mathbb{M}_D$ has no deg 3 atoms). Let $Z_{\mathbf{n}}$ be the set of ancestors of $\mathbf{n}$ in $\mathbb{M}_D$. This $\mathbf{n}$ and $Z_{\mathbf{n}}$ is fixed until all atoms in $Z_{\mathbf{n}}$ have been cut.
- (3) Starting from $\mathbf{n}$, each time choose a lowest atom $\mathbf{m}$ in $Z_{\mathbf{n}}$ that has not been cut. If $\mathbf{m}$ has deg 3 and has a parent $\mathbf{p} \in \mathbb{M}_U$ that also has deg 3, then cut $\{\mathbf{m}, \mathbf{p}\}$ as free; otherwise cut $\mathbf{m}$ as free. Repeat until all atoms in $Z_{\mathbf{n}}$ have been cut, then go to (1)–(2) and choose the next $\mathbf{n}$, and so on.
- (4) When all atoms in $\mathbb{M}_D$ have been cut, finish off $\mathbb{M}_U$ by applying the **UP** algorithm.

Note that steps (1)–(3) are essentially just **DOWN** (i.e. the dual of **UP**) applied to $\mathbb{M}_D$, except we prioritize the $\{33\}$ molecules formed by an atom in $\mathbb{M}_D$ with an atom in $\mathbb{M}_U$.

Proposition 1.21. Let $\mathbb{M}$ be toy model III. Then, after applying the cutting sequence defined in Definition 1.20, all components we get are elementary, and we have

$$\#_{\{4\}} = 1, \quad \#_{\{33\}} \geq \frac{1}{2} \cdot \#_{2\text{conn}}. \quad (1.17)$$

Proof. Since steps (1)–(3) are essentially the same as in **DOWN**, it is easy to prove all components we get are elementary and $\#_{\{4\}} = 1$, in the same way as the proof of Proposition 1.11. Now we only need to prove that $\#_{\{33\}} \geq \#_{2\text{conn}}/2$.

Let $\mathbf{n} \in \mathbb{M}_U$ be any 2-connection, which is connected to two atoms $\mathbf{m}_1, \mathbf{m}_2 \in \mathbb{M}_D$ by two bonds. Assume $\mathbf{m}_1$ is cut before $\mathbf{m}_2$ in Definition 1.20 (they cannot be cut at the same time, as Definition 1.20 (3) does not provide an option to cut a $\{33\}$ molecule with both atoms in $\mathbb{M}_D$). Arguing in the same way as in part 2 of the proof of Proposition 1.11, we know that $\mathbf{m}_2$ cannot have deg 4 when it is cut. On the other hand, note that $\mathbb{M}_D$ is a tree initially without deg 3 atoms; by arguing in the same way as in part 4 of the proof of Proposition 1.11 and calculating the quantity σ defined there but within $\mathbb{M}_D$, we conclude that $\mathbf{m}_2$ also cannot have deg 2 when it is cut.

Therefore, $\mathbf{m}_2$ must have deg 3 when it is cut; at this time $\mathbf{n}$ must also have deg 3 because $\mathbf{m}_1$ has been cut, $\mathbf{m}_2$ has not, and $\mathbf{n}$ also has two top edges (not fixed ends) that are not affected, see [Figure 20](#). By (3) we conclude that $\mathbf{m}_2$ must belong to a $\{33\}$ molecule, which means that $\mathbf{n}$ must be adjacent with an $\mathbb{M}_D$ atom in *some* $\{33\}$ molecule. Since this is true for each 2-connection, and each $\{33\}$ molecule may correspond to at most two 2-connections in this way (i.e. given the $\{33\}$ molecule there can be at most two $\mathbf{n}$ that are adjacent with the $\mathbb{M}_D$ atom in this $\{33\}$ molecule), we conclude that $\#_{\{33\}} \geq \#_{2\text{conn}}/2$, as desired. $\square$

1.5 Additions and reduction to toy models

In this subsection, we discuss the changes and additions that are necessary to address the general case, without the simplifications made in the various toy models in this section. These should be viewed as branches that are attached to the main thread of proof, which are less essential but necessary. They include the additions listed in Table 2, which will be discussed in Sections 1.5.1–1.5.4, as well as a few other ingredients which will be discussed in Section 1.5.5.

1.5.1 O-atoms and ov-segments

First, in the real problem, the molecule $\mathbb{M}$ may contain O-atoms. This leads to the more involved definition of cutting operations: now cutting an atom $\mathbf{n}$ as free introduces fixed ends not only at adjacent atoms, but also at atoms which are connected to $\mathbf{n}$ by an *ov-segment*, which is part of a particle line with all interior atoms being O-atoms. In particular each bond in this ov-segment is broken after cutting $\mathbf{n}$ as free, and a *simple pair* (formed by a free end and a fixed end) is introduced at each O-atom in this ov-segment, see Figure 21.

The notion of ov-segments is important in the analysis of O-atoms, and should be viewed as a generalization of eds/bonds. Using this, we can define the associated notions of being ov-adjacent (connected by an ov-segment), ov-parents and ov-children, and ov-components (connected components under the relation of being ov-adjacent), etc. Fortunately, in almost all cases, this generalization *does not cause any essential change* to the proof; we simply translate every sentence in the proofs in Sections 1.1–1.4 into the language of ov-segments, with “adjacent” replaced by “ov-adjacent” and “components” replaced by “ov-components”, etc. As an example, the fact ($\heartsuit$) in the proof of Lemma 1.7 now reads (Figure 22):

- ($\heartsuit\heartsuit$) Suppose we cut $S \subseteq \mathbb{M}$ as free, where S is an *ov-connected* subset of $\mathbb{M}$ and $\mathbb{M}$ is a tree. Then, for each atom $\mathbf{q} \in \mathbb{M} \setminus S$ that is connected to an atom in S by a (pre-cutting) *ov-segment* e , the post-cutting molecule $\mathbb{M}_D$ has exactly one component $X_{\mathbf{q}}$ corresponding to $\mathbf{q}$. Moreover, when viewed in $X_{\mathbf{q}}$, the operation of cutting S turns exactly one free end at $\mathbf{q}$ into a fixed end.

There are only two places where extra arguments are needed in addition to this translation, i.e. the **2CONNUP** algorithm and the definition of primitivity, which will be explained in the actual proof.

1.5.2 Reducing to two layers: layer selection

In Definition 1.1 we have assumed that $\mathbb{M}$ consists of only two layers: $\mathbb{M}_U$ and $\mathbb{M}_D$. In fact, this two-layer case already captures most of the essential difficulties in the multi-layer case, and a relatively simple process allows us to reduce the general $\mathfrak{L}$ -layer case (where $\mathfrak{L}$ is any large constant) to the 2-layer case. This is referred to as *layer selection*, which we discuss as follows.

Recall the set $r(\mathbb{M}_\ell)$ of particle lines “crossing” into layer ℓ from higher layers. We will rely on the key property of molecules $\mathbb{M}$, which asserts that each particle line in $r(\mathbb{M}_\ell)$ is *connected* to another particle line in $r(\mathbb{M}_\ell)$ via layers ℓ and below. We shall assume each layer $\mathbb{M}_\ell$ is a forest (when $\mathbb{M}_\ell$ contains cycles we would need another process called *layer refining*, see Section 1.5.3 below). The layer selection process is defined as follows, see Figure 23:

- (1) Let $s_\ell = |r(\mathbb{M}_\ell)|$ be the number of particle lines “crossing” into layer ℓ from higher layers. Then choose ℓ_U such that

$$\rho' := s_{\ell_U-1} \sim \max_{\ell'} s_{\ell'} \sim \rho, \quad \rho' \gg \max_{\ell' \geq \ell_U} s_{\ell'}. \quad (1.18)$$

Here the first condition in (1.18) states that the majority of crossings happen between layers ℓ_U and $\ell_U - 1$, and the second condition states that the crossings in layers above ℓ_U are negligible (which is useful in controlling the number of components of $\mathbb{M}_{\ell_U}$; eventually this is used to guarantee that all bonds between $\mathbb{M}_U$ and $\mathbb{M}_D$ do lead to cycles).

- (2) Consider each of the s_{ℓ_U-1} particle lines in $r(\mathbb{M}_{\ell_U})$. For $\ell_D < \ell_U$, define $\lambda_{\ell_U; \ell_D}$ to be the number of these particle lines that are connected to another such particle line (or to a cycle) *within layers between ℓ_U and ℓ_D* ; in other words, we start from layer ℓ_U and move down, keeping track of the number of particle lines that are connected to other particle lines or to a cycle *at each time*. Note that $\lambda_{\ell_U; 1} = s_{\ell_U-1}$ by the property in (1); now as in (1), we choose ℓ_D such that

$$\rho'' := \lambda_{\ell_U; \ell_D} \sim s_{\ell_U-1} = \rho', \quad \rho'' \gg \max_{\ell' > \ell_D} \lambda_{\ell_U; \ell'}. \quad (1.19)$$

- (3) Now we focus on the layers $\mathbb{M}_{[\ell_D:\ell_U]} := \cup_{\ell_D \leq \ell' \leq \ell_U} \mathbb{M}_{\ell'}$, and ignore all other layers. Consider each component Z of the set $\mathbb{M}_{(\ell_D:\ell_U)} := \cup_{\ell_D < \ell' < \ell_U} \mathbb{M}_{\ell'}$. If Z intersects at least two particle lines in $r(\mathbb{M}_{\ell_U-1})$ or contains a cycle, then by the choice of ℓ_D in (2), the number of such exceptional components is $\leq \lambda_{\ell_U;\ell_D+1} \ll \rho''$ which is negligible. We will assume this number is zero (otherwise we need an additional cutting and *pre-processing* step, see Section 1.5.5 below), then each component of $\mathbb{M}_{(\ell_D:\ell_U)}$ intersects exactly one particle line in $r(\mathbb{M}_{\ell_U-1})$ and contains no cycle. Since $\mathbb{M}_{\ell_U}$ is a forest by itself, we then see that $\mathbb{M}_{(\ell_D:\ell_U]} := \mathbb{M}_{\ell_U} \cup \mathbb{M}_{(\ell_D:\ell_U)}$ is also a forest.
- (4) We now define $\mathbb{M}_U := \mathbb{M}_{(\ell_D:\ell_U]}$ and $\mathbb{M}_D := \mathbb{M}_{\ell_D}$, which are both forests. Since we will have good control on their number of components, we may assume they are in fact trees; moreover, the definition of $\rho'' := \lambda_{\ell_U;\ell_D}$ implies that there exist $\geq \rho''$ bonds between $\mathbb{M}_U$ and $\mathbb{M}_D$ (in fact they are connected to each other in $\mathbb{M}_D$), which reduces us to the 2-layer model as in Definition 1.1.

We remark that in practice, the layer selection process is complicated by the need of layer refining (Section 1.5.3) and cutting the exceptional components in (3) (Section 1.5.5).

1.5.3 Reducing each layer to a tree: layer refining

In both the toy models in Sections 1.1–1.4 and the layer selection process in Section 1.5.2 above, it is crucial that each layer of $\mathbb{M}$ is a forest. In reality, to deal with the case where $\mathbb{M}$ contains cycles within a single layer, we need to define the *layer refining* process as follows.

Recall each single layer $\mathbb{M}_\ell$ of $\mathbb{M}$ contains finitely many C-molecules (i.e. molecules with only C-atoms) joined by O-atoms, that each C-molecule contains *at most* $\Gamma = O(1)$ independent cycles, and joining these C-molecules by O-atoms does not add any cycle. We have a simple and intuitive lemma, which states that (see Figure 24):

- For each molecule $\mathbb{M}$ with at most Γ independent cycles, we can always divide it into at most $\Gamma + 1$ subsets $\mathbb{M}_j$ ($0 \leq j \leq \Gamma$), such that each subset is a tree, and no atom in $\mathbb{M}_i$ can be parent of atom in $\mathbb{M}_j$ for $i < j$.

Now, to complete the layer refining process, we simply apply the above lemma to each of the C-molecules in each layer of $\mathbb{M}$ to get, for each $0 \leq j \leq \Gamma$, a label j subset of each such C-molecule in each layer. We then combine the label j subsets for all these C-molecules, and distribute the O-atoms appropriately, to divide each layer $\mathbb{M}_j$ into at most $\Gamma + 1$ *thin layers*, such that each thin layer is now a forest. This then allows us to repeat the layer selection process in Section 1.5.2; the use of thin layers instead of layers leads to one extra step in layer selection, but this is easily adjusted.

We remark that the above layer refining process does require the total number of independent cycles in these single-layer C-molecules to be negligible (compared to ρ). When this total number of cycles is large, we can perform an alternative cutting sequence that exploits these cycles. This requires layer refining of a subset of layers and also leads to a minor auxiliary notion of *primitivity*, which we omit here.

1.5.4 Strong and weak degeneracies

In Definition 1.1 we have ignored any degeneracies and assumed that all $\{33\}$ molecules lead to power gain. In reality there may be exceptions to this (when the (x_e, v_e) for *fixed ends* e at this $\{33\}$ molecule takes some highly specific values), where we cannot gain enough powers of ε from the integration associated with this $\{33\}$ molecule. One example is when

$$|x_{e_1} - x_{e_7}| \leq \varepsilon^{1-\kappa}, \quad |v_{e_1} - v_{e_7}| \leq \varepsilon^{1-\kappa} \quad (1.20)$$

for some small constant κ , where e_1 and e_7 are the two fixed ends of the $\{33\}$ molecule.

As such, we define a *strong degeneracy* to be an instance where (1.20) happens at two edges that *would become the two fixed ends if we cut the two atoms as a $\{33\}$ molecule* (in fact a so-called $\{33A\}$ molecule, but we ignore this minor point). Similarly we can define the notion of *weak degeneracy*, which involves one or two atoms and some conditions similar to (1.20). Note that these definitions *only involve* certain edges and atoms in $\mathbb{M}$, and is *not* a priori associated with any concrete cutting sequence.

Our strategy is such that, when there are many degeneracies, we shall perform an *alternative cutting sequence*, where we do not gain from $\{33\}$ components but instead gain from the extra degeneracy assumptions such as (1.20). Compared to the sophisticated algorithms exploiting $\{33\}$ molecules, this alternative cutting sequence essentially follows the simple procedure where we cut every atom by itself, so that most of them are cut as $\{3\}$ molecules. Usually integrating a $\{3\}$ molecule does not gain any power, but if this $\{3\}$ molecule contains an edge or atom that is involved in some degeneracy condition like (1.20), then it is clear that *this* $\{3\}$ molecule integration should gain some power, which effectively turn this $\{3\}$ molecule into a good one.

In practice, there will be two places in the overall algorithm where we need to consider degeneracies: the first time for strong degeneracies, and the second time for weak degeneracies. If there are many (strong or weak) degeneracies, then we can apply the alternative cutting sequence (with some complications as described above) to conclude; if there are few (strong and weak) degeneracies, then their effect will be negligible and we can get rid of them and proceed with the main thread of our algorithm.

1.5.5 Additional ingredients and pre-processing

Finally, we discuss a few minor ingredients that are needed in the full case, that are not discussed in Sections 1.5.1–1.5.4 above.

Cutting exceptional components. Recall the notion of exceptional components in step (3) in the layer selection process in Section 1.5.2. When there are no exceptional components, we know that $\mathbb{M}_U = \mathbb{M}_{\ell_U} \cup \mathbb{M}_{(\ell_D:\ell_U)}$ is a forest; however, when there are a (positive) negligible number of such exceptional components, we need to cut them first, before defining $\mathbb{M}_U$ which can be a forest. It is easy to show that this cutting process leads to a negligible number of bad $\{4\}$ components.

Pre-processing. In Definition 1.1, we have assumed that the 2-layer molecule $\mathbb{M}$ has no fixed ends. This is the case if there are no exceptional components in layer selection, but if there are exceptional components that are cut as above, then this will create fixed ends (and thus deg 3 atoms after we subsequently cut all deg 2 atoms) in both $\mathbb{M}_U$ and $\mathbb{M}_D$. Here, depending on the number of components of deg 3 atoms in $\mathbb{M}_D$ (similar to the dichotomy in Sections 1.2–1.3), we either apply the dual of **3COMPUP** in $\mathbb{M}_D$ similar to Section 1.3, or apply an extra *pre-processing* step to turn $\mathbb{M}_D$ into a proper molecule. This then allows us to execute the main thread algorithm, just as the case of no fixed ends.

Handling 2-connections. Recall that in the toy models, there is a case when the number of 2-connections satisfies $0 < \#_{2\text{conn}} \ll |X|$ (see Section 1.4), which is not covered in any toy model. In this case we will need to first cut all these 2-connections, so no more 2-connections remain in the resulting molecule, and then proceed as in either toy models I and I plus or toy model II, depending on the dichotomy. There will be fixed ends, but they can be handled by pre-processing as above. Finally, this also leads to some complications as the set X involved in the dichotomy needs to be replaced by some natural modifications, which will be explained in details in the algorithm **MAINUD** in the actual proof.

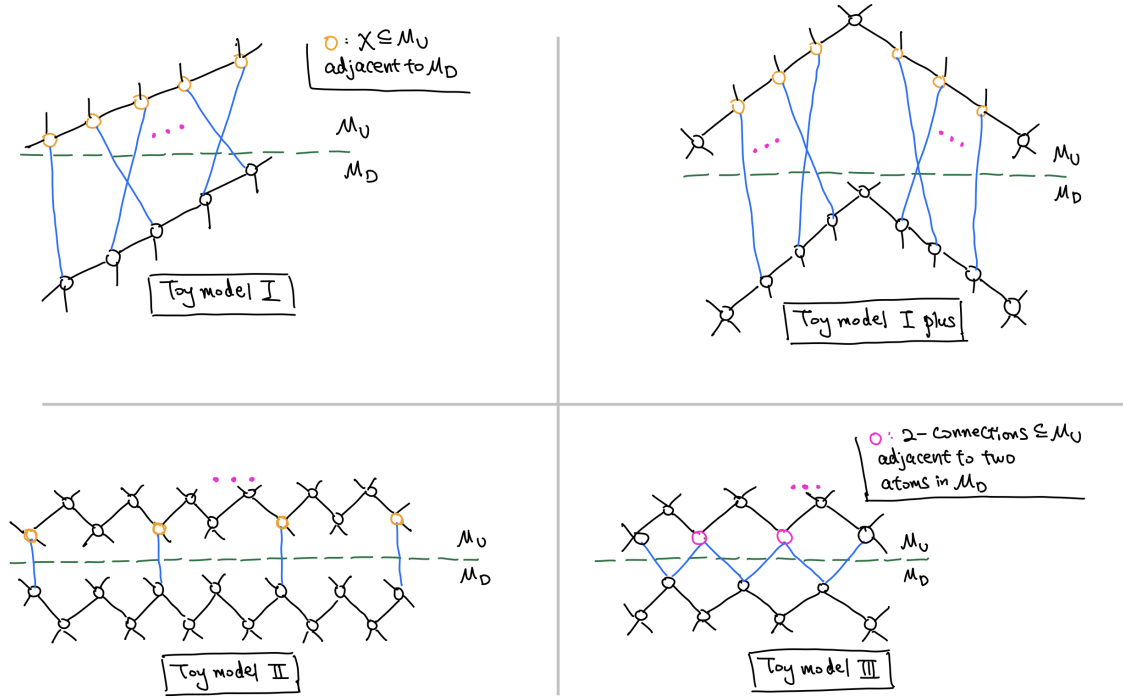


Figure 1: Examples of toy models. Upper left: toy model I ($X = \mathbb{M}_U$). Upper right: toy model I plus (where $\#_{\text{comp}}(X) = 2$). Lower left: toy model II (where $\#_{\text{comp}}(X) = |X|$). Lower right: toy model III (where $\#_{2\text{conn}} \sim |X|/2$). The blue bonds connect $\mathbb{M}_U$ and $\mathbb{M}_D$. The orange and pink atoms indicate those in X and 2-connections respectively. The pink dots represent patterns that can be repeated.

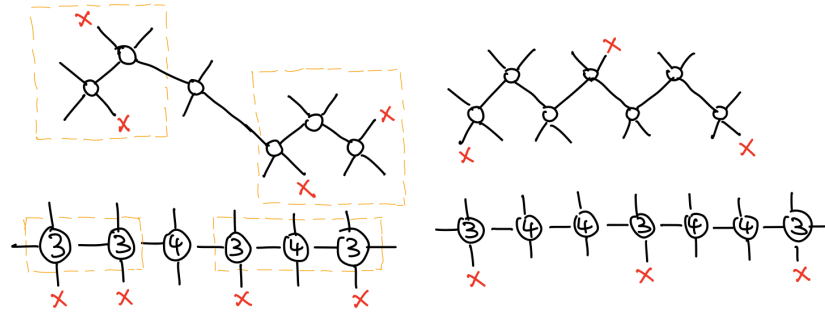


Figure 2: Examples of $\{33\}$ and $\{343\}$ molecules (left), and proper molecules (right, see Definition 1.3). Note these notions are applied to $\mathbb{M}_D$, in which the top/bottom distinction is less important, hence we draw an equivalent simplified picture below, with no distinction of top/bottom, and with the deg (i.e. degree) of each atom indicated. Similar simplified illustrations will occur a few times below.

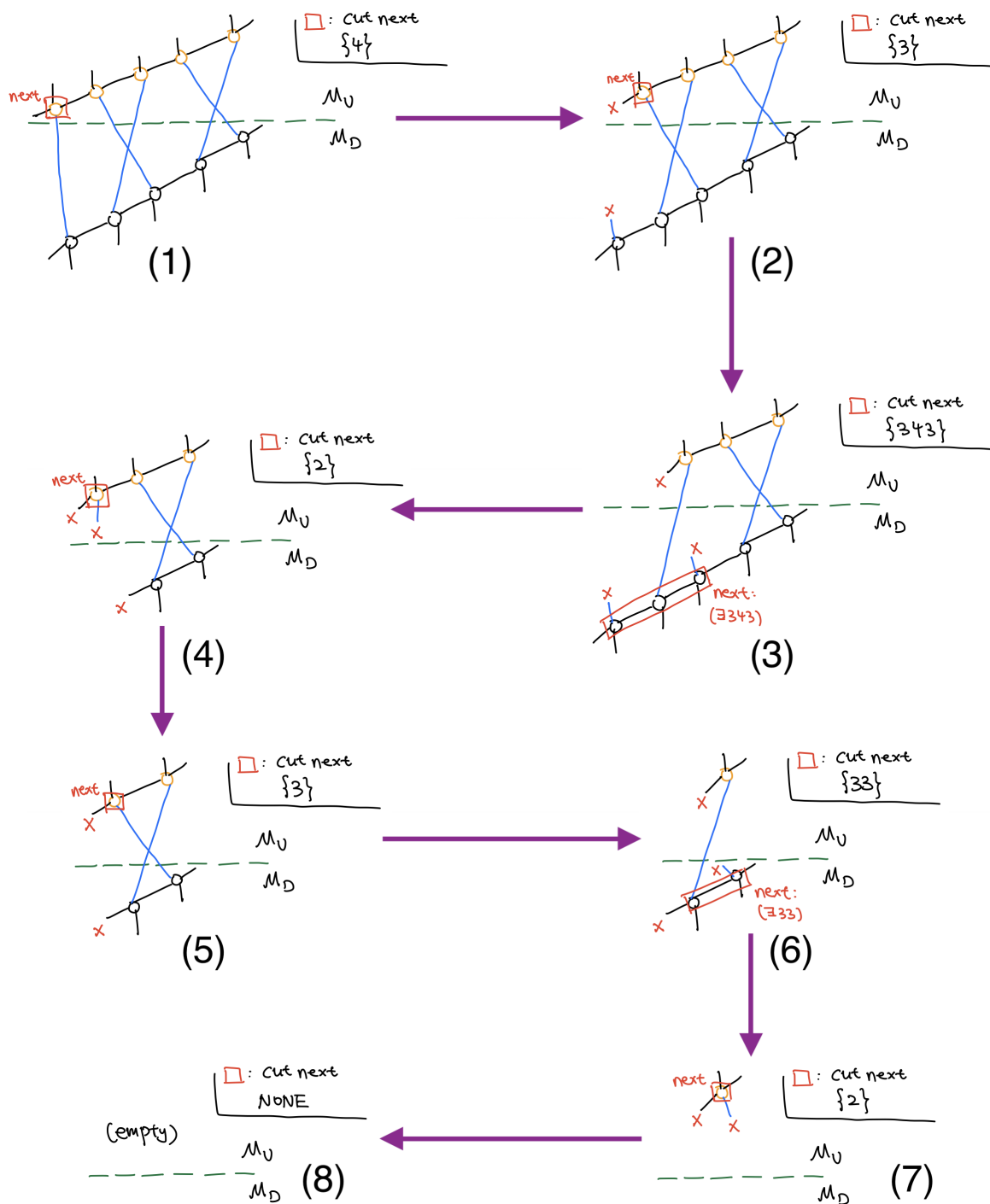


Figure 3: Algorithm for toy model I, applied to the example in Figure 1. Following the rule of the algorithm (Definition 1.4), each time we cut the atom(s) in the red box, creating fixed ends. The type of the elementary molecules to be cut is listed at the upper right corner.

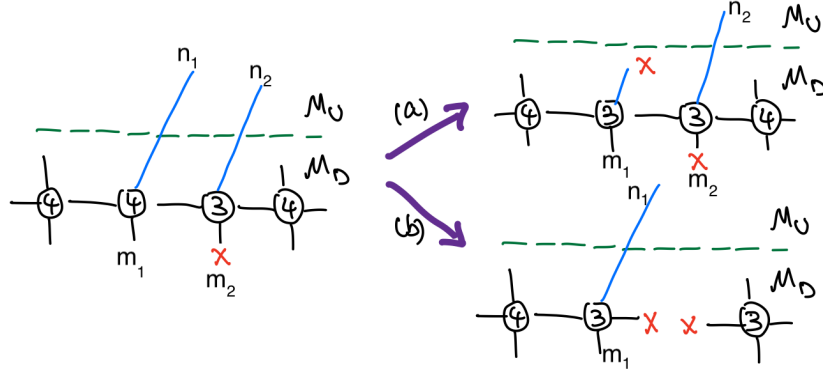


Figure 4: Operations (a) and (b) defined in the proof of Proposition 1.6. In either case an atom $m_j \in \mathbb{M}_D$ is connected to an atom $n_j \in \mathbb{M}_U$ by a blue bond. In case (a) m_1 has deg 4, and we cut $\{n_1\}$ as free, creating a fixed end at $m_1 \in \mathbb{M}_D$ and turning it into deg 3; in case (b) m_2 has deg 3, and we cut $\{n_2, m_2\}$ as free, which when viewed in $\mathbb{M}_D$ just corresponds to cutting $\{m_2\}$ as free, creating fixed ends at its adjacent atoms.

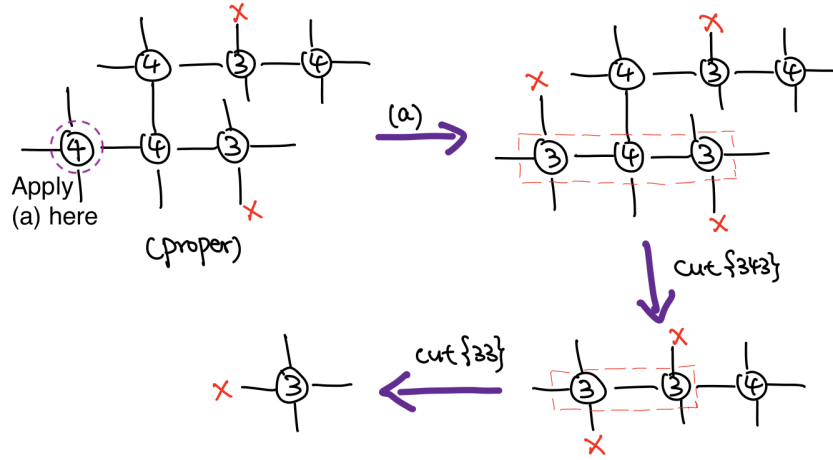


Figure 5: The greedy algorithm in Definition 1.4. Starting from upper left graph (which is proper), we apply operation (a), which creates a $\{343\}$ molecule (in red box). We cut it, which subsequently creates a $\{33\}$ molecule. We cut it, and the graph becomes proper again.

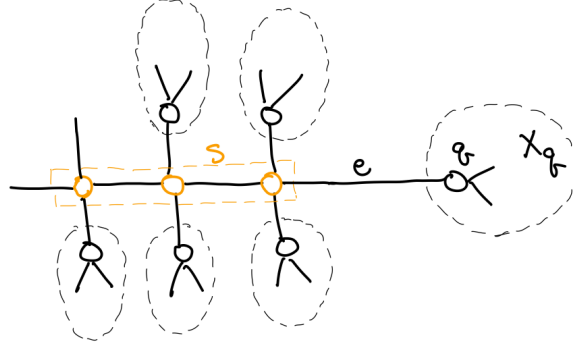


Figure 6: Property (♡) in the proof of Lemma 1.7. Here S is a connected subset, and there are bijections between $\mathfrak{q}$ and $X_{\mathfrak{q}}$, where $\mathfrak{q} \in \mathbb{M}_D \setminus S$ is connected to S by a bond e , and $X_{\mathfrak{q}}$ is the component containing $\mathfrak{q}$ after cutting S as free. When viewed in $X_{\mathfrak{q}}$, the operation of cutting S as free creates exactly one fixed end in $X_{\mathfrak{q}}$. All these crucially depend on the fact that $\mathbb{M}_U$ and $\mathbb{M}_D$ have no cycles.

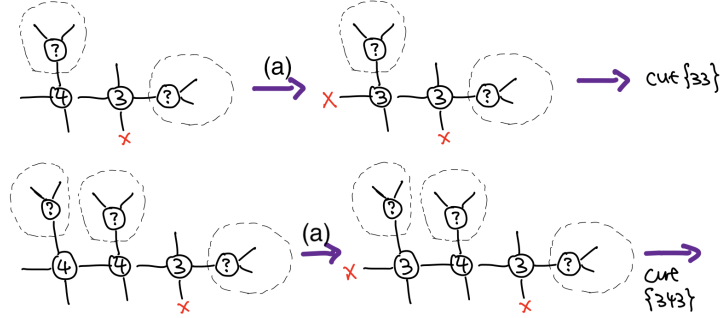


Figure 7: Proof of Lemma 1.7. After applying (a) to a proper $\mathbb{M}_D$, a deg 4 atom is turned into deg 3, thus creating a $\{33\}$ (top) or $\{343\}$ molecule (bottom). Crucially, all the atoms with “?” must be deg 4, because $\mathbb{M}_D$ is proper before applying (a), and because applying (a) does not create $\{33\}$ molecule in the bottom case. Then, after cutting the $\{33\}$ or $\{343\}$ molecule, when viewed in each resulting component (which contains one “?” atom), we create only one fixed end and turn the “?” atom from deg 4 to deg 3, and so on.

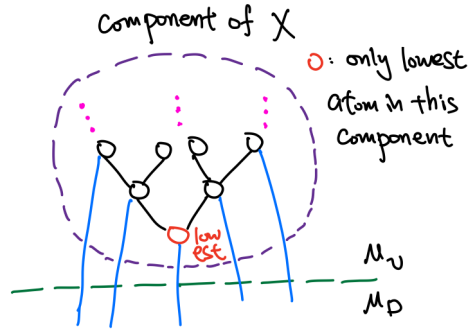


Figure 8: Illustration of each component of X (or M_U if $X = M_U$ in toy model I). It has a unique lowest atom (in red). The other atoms can only be constructed from the red one by taking *parents*, as they must have a bond connecting to M_D (in blue) and cannot have a third child.

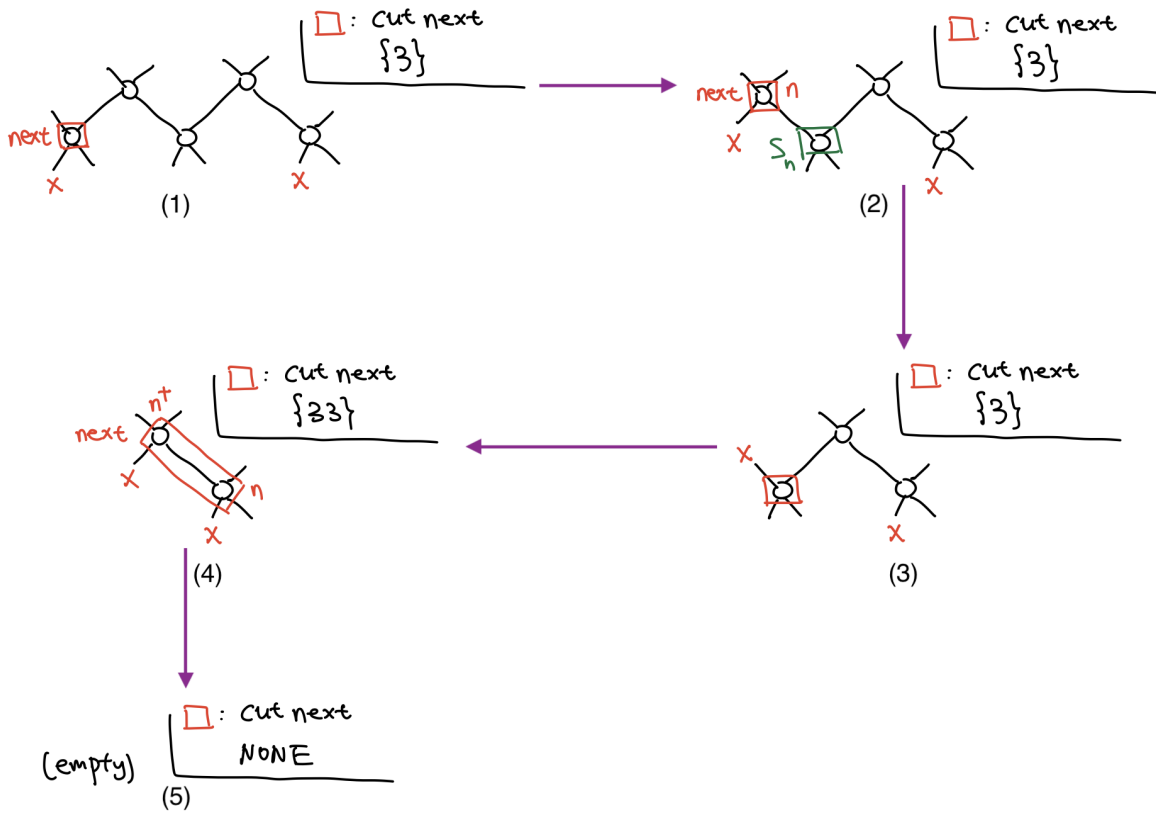


Figure 9: The **UP** algorithm. As in the above figures, the red box indicates the atom(s) to be cut next, the green box indicates the atom(s) in S_n (see Step (2) in the **UP** algorithm), which will be cut after cutting n . Note also when n and its parent n^+ (or child n^-) form a $\{33\}$ molecule, we should cut them together; see Step (3) of the **UP** algorithm.

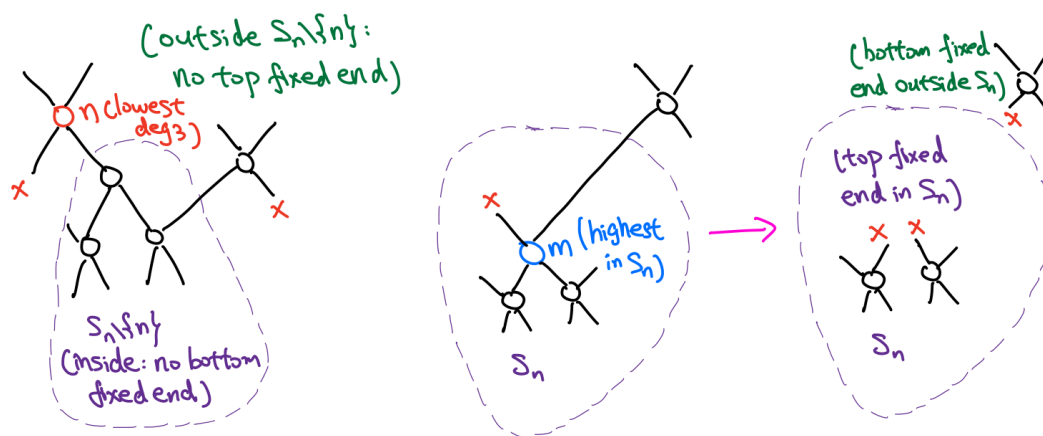


Figure 10: Illustration of the monotonicity property of the **UP** algorithm. Left: when we choose n in Step (2), there is no top fixed end outside $S_n \setminus \{n\}$ and no fixed end at all in $S_n \setminus \{n\}$ (i.e. all deg 4). Right: when we cut a highest remaining atom m in S_n , this only creates top fixed end on S_n and bottom fixed end outside S_n .

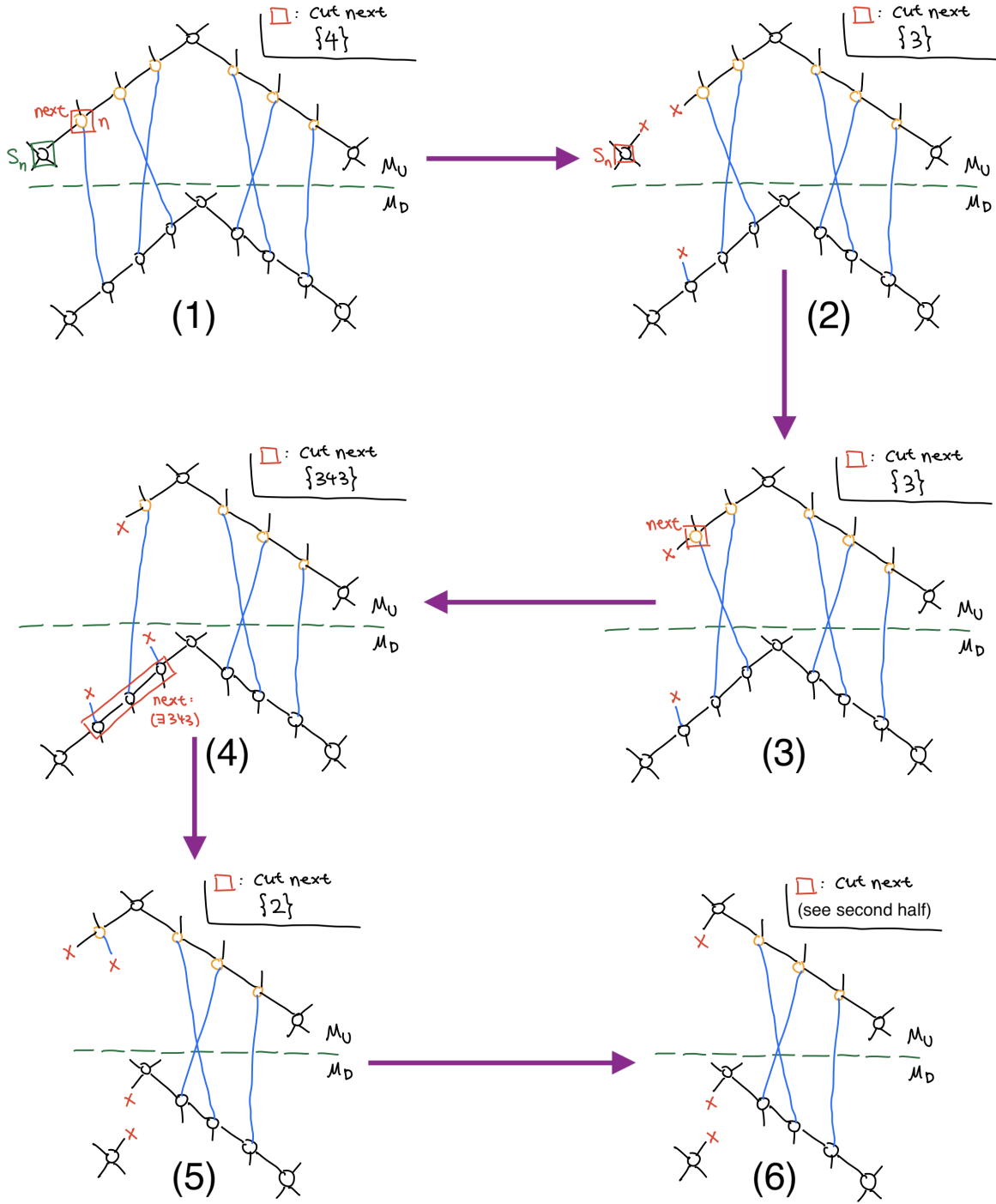


Figure 11: Algorithm for toy model I plus (first half) in Definition 1.12, applied to the example in Figure 1. In addition to Figure 3, we use green box to indicate atom(s) in S_n (see Step (2) in the toy model I plus algorithm) which will be cut after cutting n .

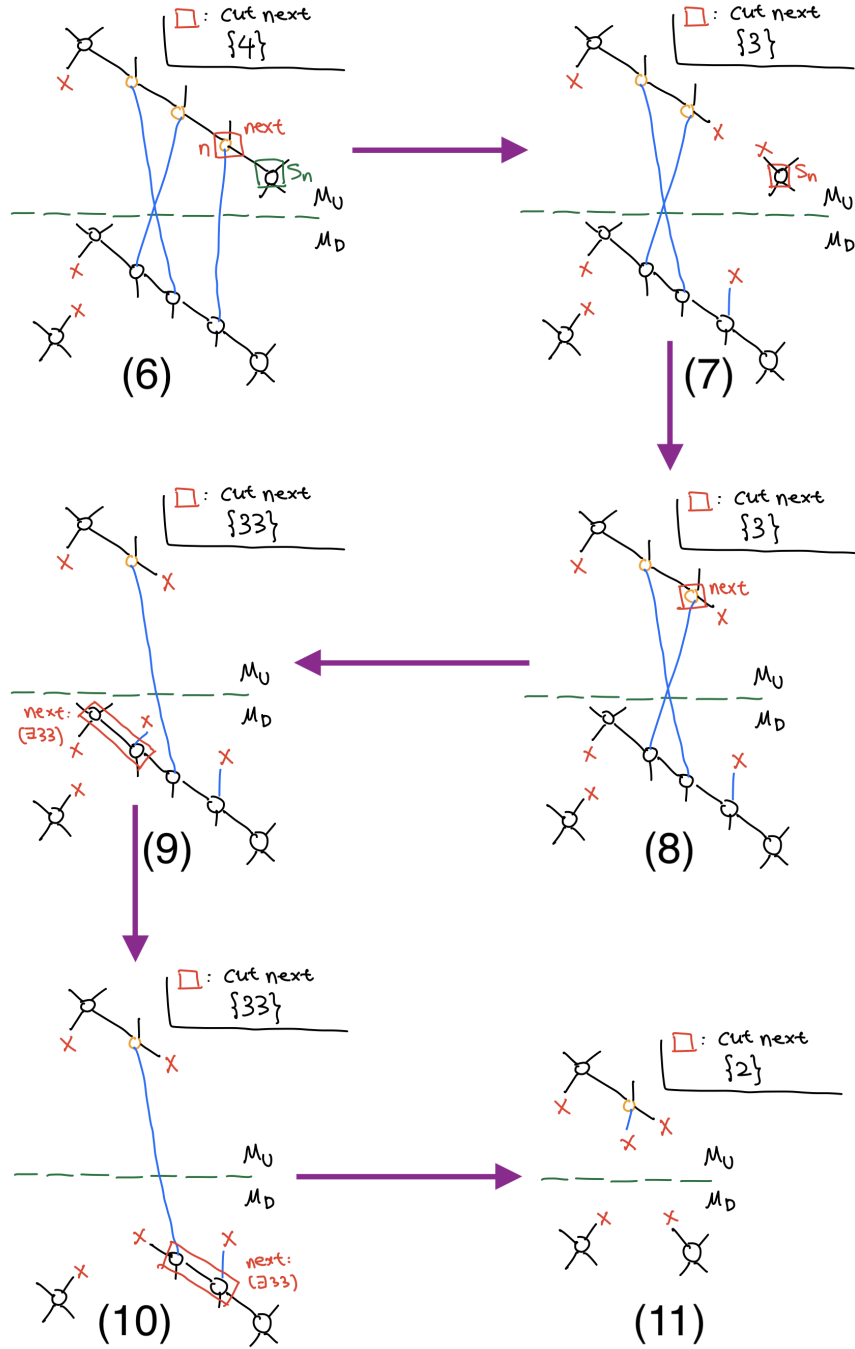


Figure 12: Algorithm for toy model I plus (second half). Here, the next atom n we choose is not the unique deg 3 atom in M_U , but rather the *atom in X* below it, see Step (2) in the toy model I plus algorithm. This leads to another $\{4\}$ loss, but the total loss is bounded by $\#_{\text{comp}(X)} = 2$. In the end, we subsequently cut two deg 2 atoms in M_U , and two deg 3 atoms in M_D , to conclude.

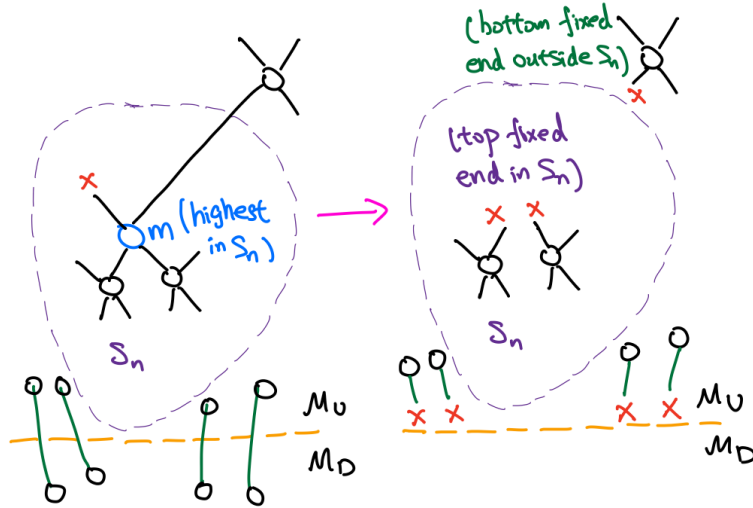


Figure 13: Part 1 of the proof of Proposition 1.13: the preservation of the monotonicity property (just as for **UP**) in $\mathbb{M}_U$. This is essentially the same as the right half of Figure 10, except the green bonds connecting $\mathbb{M}_D$ to atoms in $\mathbb{M}_U$ (**those atoms must be in X**). By our choice, $S_n \setminus \{n\}$ is **disjoint** with X , so none of these bonds involve any atom in S_n (once n has been cut). As such, if we do any cutting in $\mathbb{M}_D$ (as in Definition 1.12 (4)), this will only create bottom ends outside $S_n \setminus \{n\}$, hence preserving the monotonicity property in $\mathbb{M}_U$.

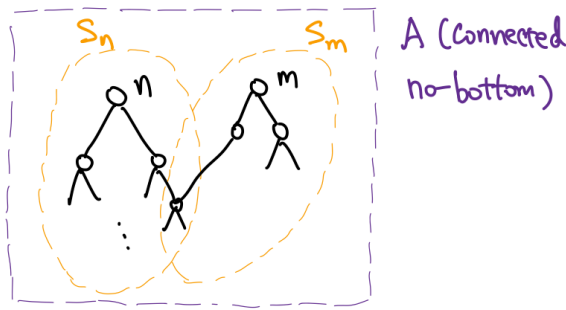


Figure 14: An example of connected, no-bottom sets in Lemma 1.15; this one is a union of two intersecting descendant sets S_m and S_n . In general each such set is a union of descendant sets S_n with n forming a connected set; see the proof of Lemma 1.15.

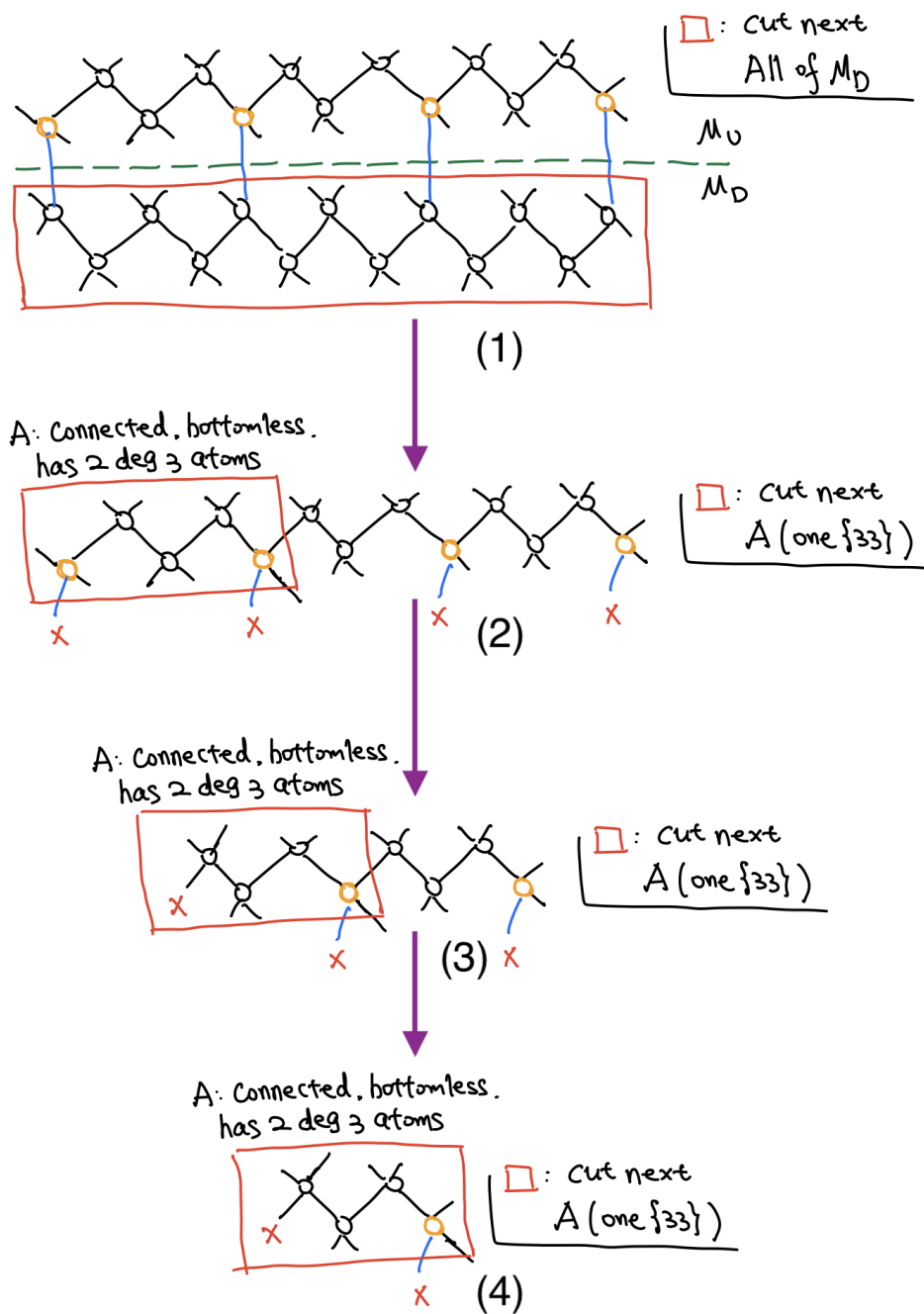


Figure 15: Algorithm for toy model II, applied to the example in Figure 1. Here we first cut all of M_D as free. Then, each time we choose a connected, no-bottom subset A that contains exactly two deg 3 atoms, see Step (1) in the toy model II algorithm. We then apply **UP** to cut all atoms in A (cf. Figure 9 for the application of **UP**) and repeat. Note that each application of **UP** guarantees at least one $\{33\}$ molecule.

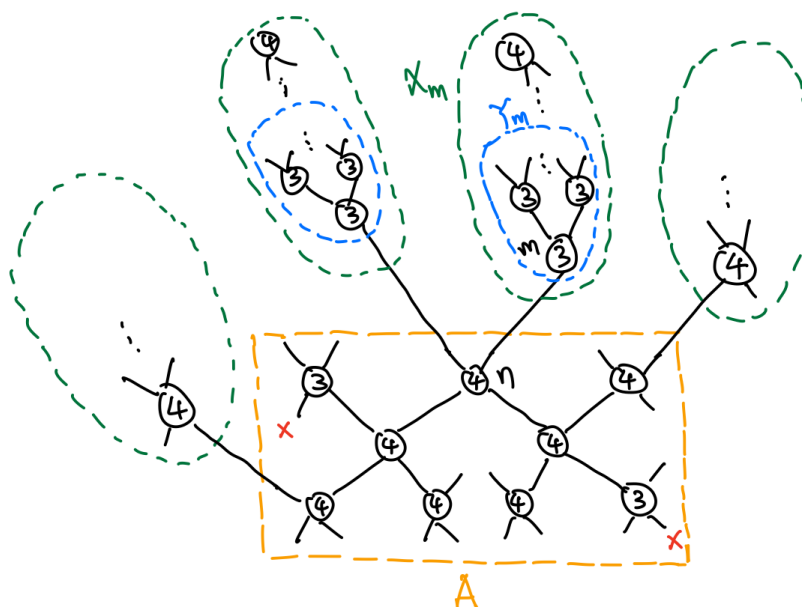


Figure 16: Part 2 of the proof of Proposition 1.18. Here A is the connected, no-bottom set as in Lemma 1.15; various atoms $\mathbf{m}$ (together with the corresponding component $X_{\mathbf{m}}$ as in ($\heartsuit$)) are drawn, each of which connected to its child $\mathbf{n} \in A$ by a bond. Apart from 2 exceptions, this $\mathbf{n}$ will have deg 4. If $\mathbf{m}$ has deg 4 we do nothing; if $\mathbf{m}$ has deg 3, we take all the subsequently connected deg 3 atoms in $X_{\mathbf{m}}$ and call them $Y_{\mathbf{m}}$ (i.e. the blue circle within the green circle $X_{\mathbf{m}}$). **The B is defined as the union of A and all blue circles.** Within one loop (1)–(5) in Definition 1.16, we cut everything in A , and subsequently everything in the blue circles as deg 2 (or as part of $\{33\}$ molecule). That is, we cut everything in B and nothing else (as cutting B only turns deg 4 atoms into deg 3 but does not produce further deg 2 atom).

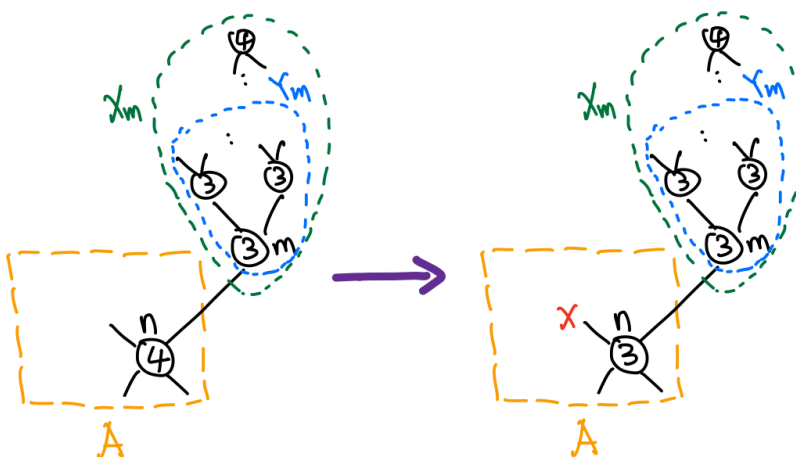


Figure 17: Part 3 of the proof of Proposition 1.18: here $\mathbf{n} \in A$ is connected to $\mathbf{m} \in X_{\mathbf{m}}$ which has deg 3, as in Figure 16. With at most 2 more exceptions, this $\mathbf{n}$ will also have deg 3 when it is cut (as we are essentially doing **UP** within A , see Proposition 1.11), which leads to a $\{33\}$ molecule formed by $\mathbf{n}$ and $\mathbf{m}$.

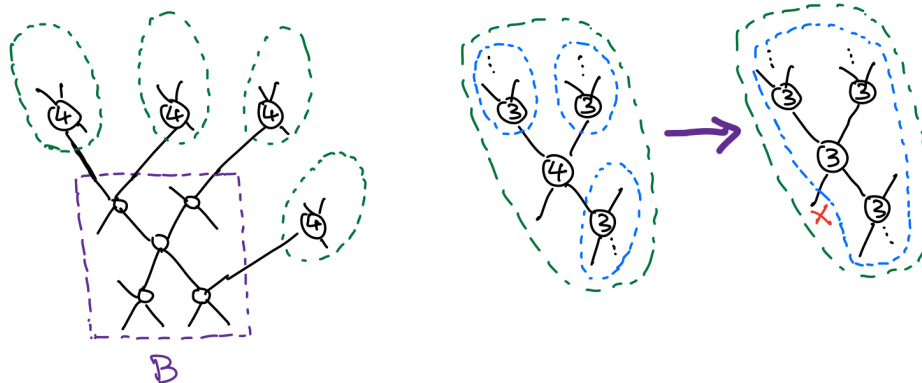


Figure 18: Part 4 of the proof of Proposition 1.18. Left: suppose we cut B and are left with all the components as in ($\heartsuit$). Within each such component, a fixed end is created, turning a deg 4 atom into deg 3. Right: for each component, turning one deg 4 atom into deg 3 may decrease the number of **components of the set of deg 3 atoms** by 2 (by joining 3 such components into 1 with the new deg 3 atom), but **no more**. Note that the blue circles here are unrelated to the blue circles in Figures 16–17 and represent different things.

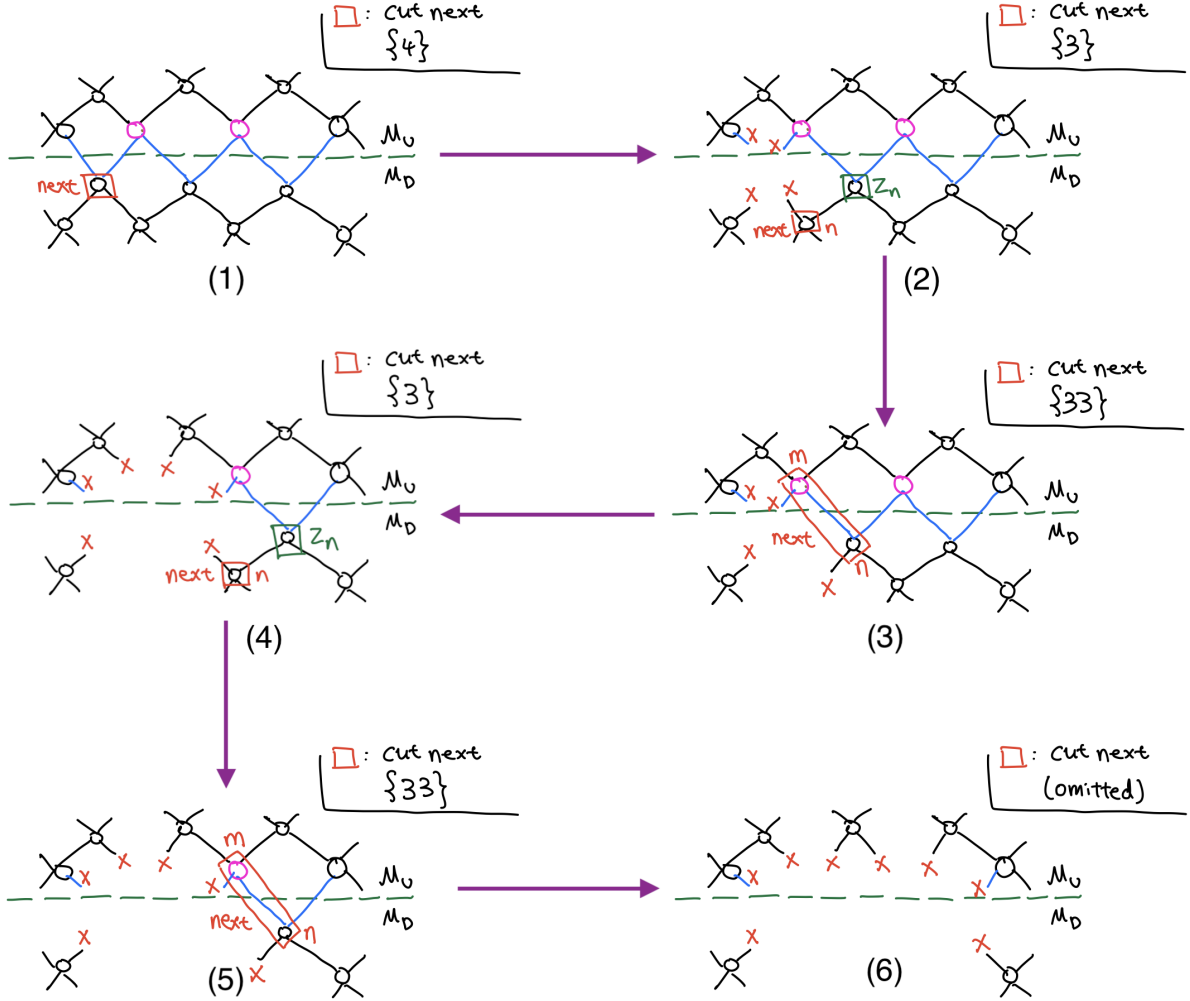


Figure 19: Algorithm for toy model III, applied to the example in Figure 1. Here we perform **DOWN** which is the dual of **UP** in $\mathbb{M}_D$, so in Step (2) in **UP** algorithm we choose the *highest* deg 3 atom instead of the lowest, and in Step (3) in **UP** algorithm we choose the *lowest* remaining atom instead of the highest, etc. Here green box indicates atoms in Z_n (the set of *ancestors* of n), and when an atom $n \in \mathbb{M}_D$ forms a $\{33\}$ molecule with an atom $m \in \mathbb{M}_U$ we always cut it. In the end we see that each 2-connection (pink atom) belongs to one $\{33\}$ molecule, as desired. As the gain is already enough, the rest can be cut into elementary molecules using **UP** in $\mathbb{M}_U$ and **DOWN** in $\mathbb{M}_D$.

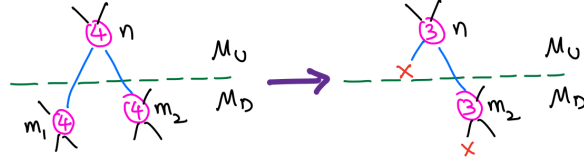


Figure 20: The proof of Proposition 1.21. Here $\mathbf{n} \in \mathbb{M}_U$ is a 2-connection connected to $\mathbf{m}_{1,2} \in \mathbb{M}_D$; assume $\mathbf{m}_1$ is cut before $\mathbf{m}_2$ when applying **DOWN** in $\mathbb{M}_D$. Then at the time $\mathbf{m}_2$ is cut, this $\mathbf{m}_2$ has deg 3 due to properties of **DOWN** (Proposition 1.11), and $\mathbf{n}$ also has deg 3 because $\mathbf{m}_1$ has been cut.

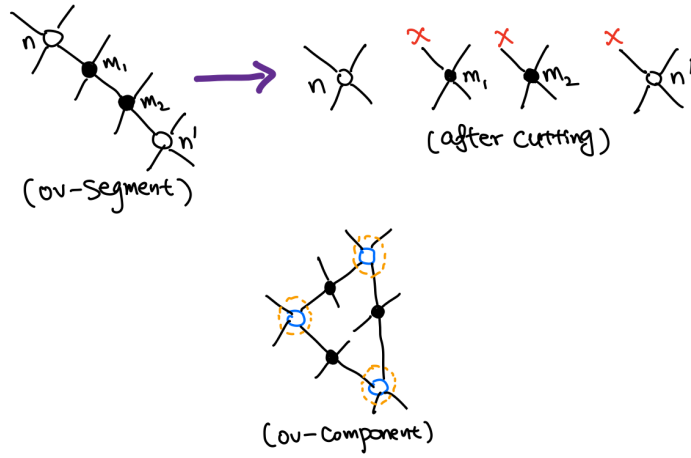


Figure 21: Notions associated with O-atoms (Section 1.5.1): *ov*-segments and associated cutting rules (which break every bond in this *ov*-segment), and *ov*-components (the 3 blue atoms in the bottom picture, which are not connected by themselves, but are so-called *ov*-connected.) Recall that hollow and solid dots represent C- and O-atoms respectively.

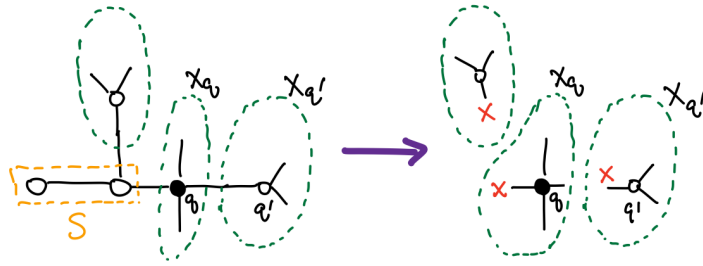


Figure 22: The property $(\heartsuit\heartsuit)$, i.e. the analog of $(\heartsuit\heartsuit)$ with O-atoms, see Section 1.5.1. Here, compared to $(\heartsuit\heartsuit)$, *each bond in each ov-segment* connecting S to S^c is broken (cf. Figure 21), so each atom $\mathbf{q}$ on this *ov*-segment will correspond to a separate component $X_{\mathbf{q}}$ (in which a new fixed end is created) after cutting S ; see for example the C-atom $\mathbf{q}$ and O-atom $\mathbf{q}'$, and the corresponding components $X_{\mathbf{q}}$ and $X_{\mathbf{q}'}$.

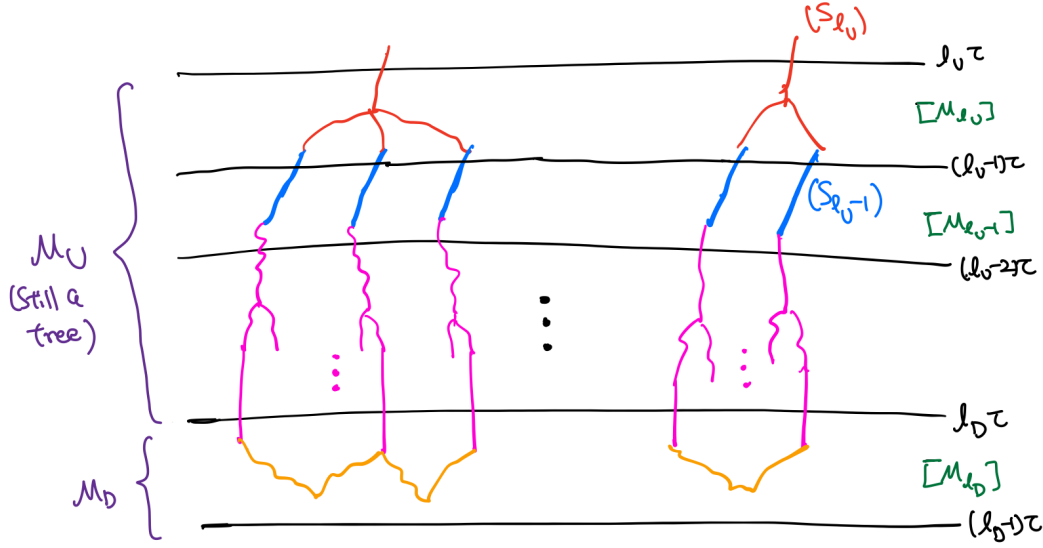


Figure 23: Layer selection (Section 1.5.2). The blue edges in layer $\mathbb{M}_{\ell_U-1}$ belong to particle lines in $r(\mathbb{M}_{\ell_U-1})$, whose size satisfies $s_{\ell_U-1} \gtrsim \rho$. Each of them is connected to a red edge crossing $\ell_U \tau$ (they belong to the particle lines in $r(\mathbb{M}_{\ell_U})$, and their number $s_{\ell_U} \ll s_{\ell_U-1}$, i.e. these particle lines are essentially connected to each other (basically, this condition is used to guarantee that all the bonds between $\mathbb{M}_U$ and $\mathbb{M}_D$ do lead to cycles). in layer $\mathbb{M}_{\ell_U}$. On the other hand, these particle lines are connected to pink edges below, but are **not connected to each other** until reaching layer $\mathbb{M}_{\ell_D}$ (by yellow edges in $\mathbb{M}_{\ell_D}$). This corresponds to the case $\lambda_{\ell_U; \ell_D+1} = 0$ and $\lambda_{\ell_U; \ell_D} \gtrsim s_{\ell_U}$ (the general case $0 < \lambda_{\ell_U; \ell_D+1} \ll s_{\ell_U}$ requires a little more work). We then set $\mathbb{M}_U$ to contain layers $\mathbb{M}_{[\ell_D+1; \ell_U]}$ (which is a forest in our case; in general case we need to cut some exceptional components to make it a forest), and $\mathbb{M}_D$ to contain layer $\mathbb{M}_{\ell_D}$, to reduce to the 2-layer case, where each of $\mathbb{M}_U$ and $\mathbb{M}_D$ contains no cycle, and the number of bonds between $\mathbb{M}_U$ and $\mathbb{M}_D$ being $\gtrsim \rho$.

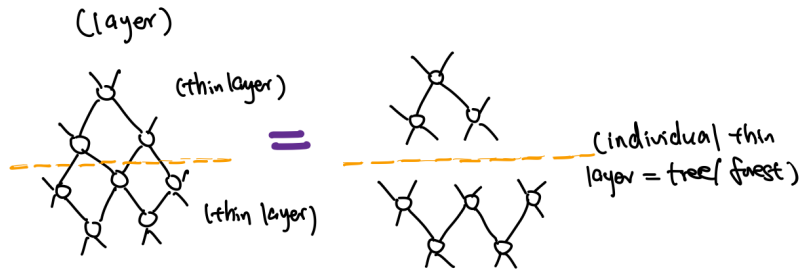


Figure 24: An illustration of layer refining (Section 1.5.3). Namely, if a single layer contains cycles, then we can break it into thin layers such that each thin layer contains no cycle. Note that here we are dealing with C-atoms only, i.e. within a C-molecule in a single layer; in practice we will do this for all C-molecules, combine the thin layers coming from different C-molecules, and distribute the O-atoms appropriately.